

Simple Studentized ℓ_2 Statistics and Gaussian Bootstrap for Two-Sample Covariance Testing

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Abstract

We propose a simple studentized ℓ_2 -type statistic for high-dimensional two-sample covariance testing. The statistic vectorizes the difference between the two sample covariance matrices, standardizes each covariance coordinate by its marginal scale, and aggregates the squared standardized discrepancies. The theoretical analysis is based on a Gaussian approximation over centered Euclidean balls, which matches the geometry of the proposed statistic and leads to a direct Gaussian multiplier-bootstrap implementation. We establish validity of the feasible bootstrap test and power consistency under growing standardized quadratic signal strength. The method is easy to implement, avoids estimating the full covariance matrix of covariance entries, and can incorporate prespecified nonsingular transformations to improve finite-sample performance.

Keywords: high-dimensional inference; Gaussian approximation; Gaussian bootstrap; multiplier bootstrap; transformed covariance statistics

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1 Introduction and Review

Covariance comparison is a fundamental problem in multivariate statistics. It arises when the scientific question concerns not only marginal variation, but also the dependence structure among variables. Examples include gene-expression studies, where genes may act jointly through co-expression modules; neuroimaging studies, where functional connectivity is encoded by covariance or correlation patterns across brain regions; finance, where portfolio risk depends on the joint movement of asset returns; and network analysis, where second-order dependence captures interaction patterns among features. In such settings, the covariance matrix is itself an object of scientific interest.

Modern covariance testing is challenging because the dimension may be comparable to, or larger than, the sample size. Classical likelihood-ratio procedures, including the likelihood criteria studied by [Box \(1949\)](#) and related covariance-matrix test criteria of [Nagao \(1973\)](#), rely on fixed-dimensional approximations and may become unstable when sample covariance matrices are singular or ill-conditioned. This difficulty motivated high-dimensional corrections for covariance structure testing, including the large-dimensional tests of [Ledoit and Wolf \(2002\)](#), as well as the nonparametric identity and sphericity tests of [Chen et al. \(2010\)](#). It has also led to a broad literature on tests that avoid direct likelihood inversion and compare covariance matrices through entrywise, spectral, or quadratic functionals.

From the perspective of power against different alternatives, many procedures are often discussed along a sparse-versus-dense contrast. Max-type tests are designed to detect alternatives where a small number of covariance entries differ sharply. This direction is connected to the limiting behavior of sample correlations and coherence statistics ([Cai and Jiang, 2011](#)), and to entrywise maximum tests for covariance comparison and support recovery ([Cai et al., 2013](#); [Chang et al., 2017](#)). Its modern calibration is closely related to high-dimensional Gaussian approximation and multiplier bootstrap theory for maxima and

hyperrectangles (Chernozhukov et al., 2013, 2017, 2023). In contrast, ℓ_2 -type procedures aggregate discrepancies across many covariance coordinates and are therefore natural for dense or moderately dense alternatives. Early and influential quadratic-form tests include those of Schott (2007) and Srivastava and Yanagihara (2010), while Li and Chen (2012) proposed an unbiased U-statistic estimator of the squared Frobenius-norm covariance difference. More recent work has further developed normal-reference and quadratic-form approaches for high-dimensional covariance testing (Wang et al., 2024).

U-statistics play an important role in this literature because they provide unbiased estimators of quadratic functionals and remove the leading bias that appears in naive plug-in estimators such as $\|\widehat{\Sigma}_1 - \widehat{\Sigma}_2\|_F^2$. The Li–Chen statistic is a canonical example in two-sample covariance testing (Li and Chen, 2012), and its debiasing idea is related to high-dimensional two-sample mean testing based on quadratic U-statistics (Chen and Qin, 2010). More generally, high-dimensional Gaussian and bootstrap approximations for non-degenerate U-statistics have been studied by Chen (2018), with related exchangeable-pair bounds by Cheng et al. (2022). These results are highly relevant, but they do not directly yield the approximation needed in the present paper. Standard U-statistic asymptotics often distinguishes non-degenerate kernels, where a leading Hájek projection drives the limit, from degenerate kernels, where higher-order components can become central. Our statistic has a different geometric target.

To see the distinction, consider the oracle version of our statistic under the null. After diagonal standardization, it can be written as the squared Euclidean norm of a high-dimensional covariance-coordinate mean vector. If one expands this squared norm, the off-diagonal part has a degenerate second-order-kernel form, while the diagonal part and the studentization remain essential to the statistic. Thus a direct U-statistic normal approximation is not the right object: the null distribution is naturally described as a quadratic form of

a Gaussian vector, or equivalently through probabilities of centered Euclidean balls. This is the main reason we develop the theory through a centered-ball Gaussian approximation for the covariance-coordinate vector, rather than through a scalar U-statistic central limit theorem.

We propose a simple studentized ℓ_2 -type statistic for two-sample covariance testing. The statistic vectorizes the upper-triangular entries of the two sample covariance matrices, forms their two-sample difference, standardizes each covariance coordinate by its marginal scale, and aggregates the squared standardized discrepancies. This produces a scale-invariant global measure of covariance difference. The construction is particularly suited to dense alternatives, where many covariance entries change moderately rather than a few entries changing sharply.

The key theoretical idea is to analyze the standardized covariance-coordinate vector before applying the squared-norm map. Events of the form

$$\{d^{-1}\|Z\|_2^2 \leq t\}$$

are exactly centered Euclidean balls; see the statistic in (6) and its acceptance-region representation in (7). Therefore a Gaussian approximation over centered balls matches the geometry of the proposed statistic. This approach leads to a direct multiplier-bootstrap calibration: the bootstrap is applied to the same covariance-coordinate contributions used in the statistic, and only a diagonal studentizer is needed. In particular, the procedure avoids estimating or inverting the full covariance matrix of covariance entries.

Our contributions are as follows. First, we introduce a simple and interpretable studentized ℓ_2 statistic for high-dimensional two-sample covariance testing. Second, we establish a Gaussian approximation over centered Euclidean balls for the oracle statistic, with explicit dependence on dimension, moments, and covariance geometry. Third, we prove that the feasible statistic, based on empirical centering and empirical diagonal studentization, is

close to its oracle counterpart. Fourth, we develop a multiplier-bootstrap calibration and prove asymptotic size validity under explicit sufficient conditions. Fifth, we derive power consistency under alternatives with growing standardized quadratic signal strength. The framework can also incorporate common nonsingular transformations, allowing external scientific information or algorithmic representations to make covariance differences more visible after studentization. The formal theory treats such transformations as fixed and deterministic, while fully adaptive same-sample selection is left as a separate problem.

The remainder of the paper is organized as follows. Section 2 presents the problem setting, the proposed studentized statistic, the centered-ball Gaussian approximation, the sample-to-oracle comparison, bootstrap calibration, size control, and power analysis. Section 3 reports simulation studies comparing the proposed bootstrap test with max-type and quadratic-form benchmarks. Section 4 presents an application to TCGA-BRCA gene-expression data. Section 5 concludes with discussion and open directions.

2 Main Results

This section develops the proposed studentized two-sample covariance test under a deterministic linear transformation. We first introduce the testing problem (1), the null-preserving transformation (2), and the statistic (6). We then establish the oracle Gaussian approximation, the sample-to-oracle transfer, the multiplier-bootstrap calibration, size validity, and power consistency. We close the section by highlighting how the simplicity of the proposed quadratic form leads to a transparent inferential interpretation. In highly unbalanced designs, the statistic can be naturally viewed as a one-sample covariance discrepancy, with the larger sample providing a stable reference covariance for assessing the smaller sample.

2.1 Problem setting

Let

$$X_1, \dots, X_{n_1} \in \mathbb{R}^p, \quad Y_1, \dots, Y_{n_2} \in \mathbb{R}^p$$

be two independent samples, with observations independent and identically distributed within each sample. Denote

$$\mathbb{E}X_i = \mu_1, \quad \text{Cov}(X_i) = \Sigma_1,$$

and

$$\mathbb{E}Y_j = \mu_2, \quad \text{Cov}(Y_j) = \Sigma_2.$$

We consider the two-sample covariance testing problem

$$H_0 : \Sigma_1 = \Sigma_2 \quad \text{against} \quad H_1 : \Sigma_1 \neq \Sigma_2. \quad (1)$$

Throughout the theoretical analysis, $A \in \mathbb{R}^{p \times p}$ is treated as a deterministic nonsingular matrix. Define the transformed observations

$$X_i^{(A)} = AX_i, \quad Y_j^{(A)} = AY_j,$$

with transformed covariance matrices

$$\Sigma_1^{(A)} = A\Sigma_1A^\top, \quad \Sigma_2^{(A)} = A\Sigma_2A^\top.$$

Since A is nonsingular,

$$\Sigma_1 = \Sigma_2 \iff \Sigma_1^{(A)} = \Sigma_2^{(A)}. \quad (2)$$

Thus the equivalence in (2) shows that applying the same deterministic transformation to both samples preserves the null hypothesis. The transformation changes the geometric representation of the covariance difference but not the null itself.

The matrix A should therefore be viewed as an analysis-side representation choice rather than as a modification of the hypothesis. The baseline choice $A = I_p$ tests covariance

equality in the original coordinates. Other nonsingular choices can encode external scientific knowledge, pre-existing feature maps, or transformations suggested by machine-learning preprocessing. In this sense, A offers a representation-augmentation mechanism: the same null-preserving test can be applied after re-expressing the data in coordinates where the anticipated covariance discrepancy may be denser, better scaled, or more aligned with the ℓ_2 aggregation. The validity results below are stated for fixed deterministic A . If A is estimated or selected from the same observations used for testing, additional sample-splitting, conditioning, or post-selection analysis would be required.

After introducing the transformation, we suppress the superscript (A) and work entirely in the transformed model. Thus $X_i, Y_j, \mu_1, \mu_2, \Sigma_1$, and Σ_2 below denote their transformed counterparts. Let

$$d = \frac{p(p+1)}{2}, \quad n_{\text{eff}} = \frac{n_1 n_2}{n_1 + n_2}. \quad (3)$$

Here d is the number of upper-triangular covariance coordinates, and $\text{vech}(\cdot)$ denotes the half-vectorization operator that stacks the upper-triangular entries of a symmetric matrix, including the diagonal. The effective sample size n_{eff} in (3) is the scale that appears in the covariance difference statistic below.

We impose the following primitive regularity conditions in the transformed coordinate system. Throughout the theoretical analysis, write

$$U_i = X_i - \mu_1, \quad V_j = Y_j - \mu_2.$$

Assumption 2.1 (Coordinate high-moment control). There exists a constant $K_8 < \infty$, independent of n_1, n_2 , and p , such that

$$\max_{1 \leq a \leq p} \mathbb{E}|U_{ia}|^8 + \max_{1 \leq a \leq p} \mathbb{E}|V_{ja}|^8 \leq K_8.$$

Assumption 2.2 (Covariance-coordinate nondegeneracy). There exists a constant $c_H > 0$, independent of n_1, n_2 , and p , such that for every $1 \leq a \leq b \leq p$,

$$\text{Var}(U_{ia}U_{ib}) \geq c_H, \quad \text{Var}(V_{ja}V_{jb}) \geq c_H.$$

Assumption 2.1 is a coordinatewise moment condition used to control covariance-coordinate estimators uniformly over the $d = p(p+1)/2$ entries. It also implies the Euclidean moment and sample-mean bounds needed for sample-centering arguments; the short derivation is given in the supplement. Assumption 2.2 excludes degenerate covariance coordinates, so that the diagonal studentizer is well defined and stable. These assumptions are primitive because they are stated directly in terms of transformed observations and covariance-coordinate variances, rather than in terms of the limiting statistic itself.

2.2 Statistic and Gaussian approximation

The sample means and sample covariance matrices are

$$\bar{X} = \frac{1}{n_1} \sum_{i=1}^{n_1} X_i, \quad \bar{Y} = \frac{1}{n_2} \sum_{j=1}^{n_2} Y_j,$$

and

$$\begin{aligned} \widehat{\Sigma}_1 &= \frac{1}{n_1} \sum_{i=1}^{n_1} (X_i - \bar{X})(X_i - \bar{X})^\top, \\ \widehat{\Sigma}_2 &= \frac{1}{n_2} \sum_{j=1}^{n_2} (Y_j - \bar{Y})(Y_j - \bar{Y})^\top. \end{aligned}$$

Define the scaled covariance-difference vector

$$\widehat{W} = \sqrt{n_{\text{eff}}} \text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_2) \in \mathbb{R}^d. \quad (4)$$

For $k = k(a, b)$, $1 \leq a \leq b \leq p$, define

$$\widehat{h}_{i,k}^{(1)} = (X_{ia} - \bar{X}_a)(X_{ib} - \bar{X}_b) - \widehat{\Sigma}_{1,ab},$$

and

$$\hat{h}_{j,k}^{(2)} = (Y_{ja} - \bar{Y}_a)(Y_{jb} - \bar{Y}_b) - \widehat{\Sigma}_{2,ab}.$$

The diagonal variance estimator is

$$\widehat{D}_{kk} = n_{\text{eff}} \left(\frac{1}{n_1^2} \sum_{i=1}^{n_1} (\hat{h}_{i,k}^{(1)})^2 + \frac{1}{n_2^2} \sum_{j=1}^{n_2} (\hat{h}_{j,k}^{(2)})^2 \right), \quad 1 \leq k \leq d. \quad (5)$$

Let

$$\widehat{D} = \text{diag}(\widehat{D}_{11}, \dots, \widehat{D}_{dd}).$$

The proposed statistic is

$$\widehat{T}_n = \frac{1}{d} \widehat{W}^\top \widehat{D}^{-1} \widehat{W} = \frac{1}{d} \left\| \widehat{D}^{-1/2} \widehat{W} \right\|_2^2. \quad (6)$$

Large values of $\widehat{T}_n$ in (6) provide evidence against H_0 .

The statistic is an ℓ_2 -type aggregation of standardized covariance differences based on the scaled difference (4) and diagonal studentizer (5). Its acceptance region satisfies

$$\{\widehat{T}_n \leq t\} = \left\{ \left\| \widehat{D}^{-1/2} \widehat{W} \right\|_2 \leq \sqrt{dt} \right\}. \quad (7)$$

Thus, after diagonal standardization, the relevant geometry is Euclidean-ball geometry.

For theoretical analysis, define the oracle covariance estimators

$$\widetilde{\Sigma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} (X_i - \mu_1)(X_i - \mu_1)^\top,$$

$$\widetilde{\Sigma}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} (Y_j - \mu_2)(Y_j - \mu_2)^\top,$$

and

$$\widetilde{W} = \sqrt{n_{\text{eff}}} \text{vech}(\widetilde{\Sigma}_1 - \widetilde{\Sigma}_2).$$

It is useful to introduce the associated covariance-coordinate notation. For $g = 1, 2$, define

$$H_i^{(1)} = \text{vech}\{U_i U_i^\top - \Sigma_1\}, \quad H_j^{(2)} = \text{vech}\{V_j V_j^\top - \Sigma_2\},$$

and write

$$\Gamma_g = \text{Cov}(H^{(g)}), \quad D_g = \text{diag}(\Gamma_g),$$

where $H^{(1)}$ and $H^{(2)}$ denote generic copies of $H_i^{(1)}$ and $H_j^{(2)}$, respectively. Also set

$$\bar{H}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} H_i^{(1)}, \quad \bar{H}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} H_j^{(2)}.$$

Under H_0 ,

$$\widetilde{W} = \sqrt{n_{\text{eff}}}(\bar{H}_1 - \bar{H}_2).$$

Let

$$\widetilde{D} = \text{diag}\{\text{Var}(\widetilde{W}_1), \dots, \text{Var}(\widetilde{W}_d)\}.$$

The oracle statistic is

$$\widetilde{T}_n = \frac{1}{d} \|\widetilde{D}^{-1/2} \widetilde{W}\|_2^2. \quad (8)$$

Under H_0 , the standardized vector

$$\widetilde{D}^{-1/2} \widetilde{W}$$

is a sum of independent centered random vectors. We collect its covariance matrix and the associated oracle Gaussian reference statistic as

$$\begin{aligned} R &= \widetilde{D}^{-1/2} \text{Cov}(\widetilde{W}) \widetilde{D}^{-1/2}, \\ T_Z &= \frac{1}{d} \|Z\|_2^2, \quad Z \sim N(0, R). \end{aligned} \quad (9)$$

Accordingly, (9) defines the oracle Gaussian reference statistic associated with the oracle statistic (8).

Let

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$$

be the eigenvalues of R . For $s = 1, 2$, define

$$\Lambda_s = \left(\sum_{\ell=s}^d \lambda_\ell^2 \right)^{1/2}, \quad \kappa(R) = (\Lambda_1 \Lambda_2)^{-1/2} \quad \text{whenever } \Lambda_2 > 0. \quad (10)$$

Assumption 2.3 (Gaussian reference geometry). The oracle Gaussian reference matrix R satisfies

$$\Lambda_2 > 0.$$

Assumption 2.3 is a mild geometric condition on the Gaussian quadratic reference, expressed through (10). It rules out the degenerate case in which the reference variation is concentrated in a single direction, where anti-concentration for Euclidean balls becomes ill-posed.

The first theorem establishes that the oracle statistic (8) is approximated by the Gaussian quadratic reference (9).

Theorem 2.1 (Oracle Gaussian approximation). *Suppose $H_0 : \Sigma_1 = \Sigma_2$ holds. Suppose Assumptions 2.1, 2.2, and 2.3 hold. Then there exists a constant $K > 0$, depending only on the moment constants and the nondegeneracy constant in the assumptions, such that*

$$\sup_{r \geq 0} \left| \mathbb{P}(\tilde{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq K \left\{ \kappa(R)^{3/4} \delta_1^{1/4} + \kappa(R)^{2/3} \delta_2^{1/3} \right\},$$

where

$$\delta_1 = \|R\|_{\text{HS}} d^2 (n_1^{-1} + n_2^{-1}) + \|R\|_{\text{op}}^{3/2} d^{3/2} (n_1^{-1/2} + n_2^{-1/2}),$$

and

$$\delta_2 = \|R\|_{\text{op}}^{1/2} d^{3/2} (n_1^{-1/2} + n_2^{-1/2}) + d^2 (n_1^{-1} + n_2^{-1}).$$

The oracle approximation in Theorem 2.1 is stated for $\tilde{T}_n$, which depends on the unknown population means and the oracle diagonal variance matrix. We next show that the feasible statistic $\hat{T}_n$, obtained by replacing the population means and the oracle studentizer by their empirical counterparts, remains close to $\tilde{T}_n$. The proof relies on two ingredients. The first is the sample-centering identity

$$\hat{\Sigma}_m - \tilde{\Sigma}_m = -(\bar{X}_m - \mu_m)(\bar{X}_m - \mu_m)^\top, \quad m = 1, 2,$$

where $(\bar{X}_1, \mu_1) = (\bar{X}, \mu_1)$ and $(\bar{X}_2, \mu_2) = (\bar{Y}, \mu_2)$. The second is the uniform consistency of the diagonal studentizer:

$$\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right| = O_{\mathbb{P}} \left(\sqrt{\frac{d}{n_0}} \right), \quad n_0 = n_1 \wedge n_2.$$

This studentizer consistency follows from the primitive moment and covariance-coordinate nondegeneracy assumptions; its technical verification is deferred to the appendix.

For $\eta \in (0, 1)$, define

$$B_T(\eta) = K_0 \eta^{-1/4},$$

$$B_W(\eta) = K_0 \frac{\sqrt{n_{\text{eff}}}}{\sqrt{d}} \eta^{-1/2} \left(\frac{p}{n_1} + \frac{p}{n_2} \right),$$

and

$$B_D(\eta) = K_0 \left(\frac{d}{n_0 \eta} \right)^{1/2}, \quad n_0 = n_1 \wedge n_2,$$

where $K_0 > 0$ denotes a generic constant depending only on the constants in the assumptions. Set

$$\Delta(\eta) = 2B_D(\eta)\{B_T(\eta) + B_W(\eta)\}^2 + 2B_T(\eta)B_W(\eta) + B_W^2(\eta).$$

Proposition 2.1 (Consistency of the feasible statistic). Suppose $H_0 : \Sigma_1 = \Sigma_2$ holds. Suppose Assumptions 2.1 and 2.2 hold. If $B_D(\eta) < 1/2$, then with probability at least $1 - \eta$,

$$|\widehat{T}_n - \widetilde{T}_n| \leq \Delta(\eta).$$

In particular, taking

$$\eta = n_0^{-1/8},$$

we have, with probability at least $1 - n_0^{-1/8}$,

$$|\widehat{T}_n - \widetilde{T}_n| \leq K_0 \left(\frac{d}{n_0^{3/4}} \right)^{1/2}.$$

Combining Theorem 2.1 with Proposition 2.1 gives a Gaussian approximation for the feasible statistic. Let

$$\varepsilon_{\text{or}} = K \left\{ \kappa(R)^{3/4} \delta_1^{1/4} + \kappa(R)^{2/3} \delta_2^{1/3} \right\}.$$

Define the feasible approximation error

$$\varepsilon_{\text{feas}}(\eta) = \varepsilon_{\text{or}} + \eta + K \kappa(R) d \Delta(\eta). \quad (11)$$

Corollary 2.2 (Feasible Gaussian approximation). *Suppose $H_0 : \Sigma_1 = \Sigma_2$ holds. Suppose Assumptions 2.1, 2.2, and 2.3 hold. If*

$$B_D(\eta) \leq \frac{1}{2},$$

then

$$\sup_{r \geq 0} \left| \mathbb{P}(\widehat{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq \varepsilon_{\text{feas}}(\eta).$$

2.3 Bootstrap calibration

The oracle Gaussian reference depends on the unknown covariance matrix R . We therefore calibrate the test by a Gaussian multiplier bootstrap. Let

$$e_1, \dots, e_{n_1}, \quad e'_1, \dots, e'_{n_2}$$

be i.i.d. $N(0, 1)$ variables independent of the data. Define

$$\widehat{h}_i^{(1)} = (\widehat{h}_{i,1}^{(1)}, \dots, \widehat{h}_{i,d}^{(1)})^\top, \quad \widehat{h}_j^{(2)} = (\widehat{h}_{j,1}^{(2)}, \dots, \widehat{h}_{j,d}^{(2)})^\top.$$

The multiplier bootstrap vector is

$$\widehat{W}^{\text{mb}} = \sqrt{n_{\text{eff}}} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} e_i \widehat{h}_i^{(1)} - \frac{1}{n_2} \sum_{j=1}^{n_2} e'_j \widehat{h}_j^{(2)} \right\}. \quad (12)$$

The bootstrap statistic is

$$\widehat{T}_n^{\text{mb}} = \frac{1}{d} \left\| \widehat{D}^{-1/2} \widehat{W}^{\text{mb}} \right\|_2^2. \quad (13)$$

Conditionally on the data,

$$\widehat{D}^{-1/2}\widehat{W}^{\text{mb}} \sim N(0, \widehat{R}),$$

where

$$\widehat{R} = \widehat{D}^{-1/2}\widehat{\Gamma}\widehat{D}^{-1/2},$$

and

$$\widehat{\Gamma} = n_{\text{eff}} \left\{ \frac{1}{n_1^2} \sum_{i=1}^{n_1} \widehat{h}_i^{(1)} (\widehat{h}_i^{(1)})^\top + \frac{1}{n_2^2} \sum_{j=1}^{n_2} \widehat{h}_j^{(2)} (\widehat{h}_j^{(2)})^\top \right\}.$$

The bootstrap critical value is

$$\widehat{c}_{1-\alpha}^{\text{mb}} = \inf \left\{ t \geq 0 : \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \leq t \mid \mathcal{X} \right) \geq 1 - \alpha \right\}. \quad (14)$$

The level- α test rejects H_0 when

$$\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}. \quad (15)$$

In implementation, the conditional quantile is approximated by Monte Carlo simulation. For $b = 1, \dots, B$, generate independent Gaussian multipliers, compute the bootstrap statistic in (13), and take the empirical $(1 - \alpha)$ -quantile of

$$\widehat{T}_{n,1}^{\text{mb}}, \dots, \widehat{T}_{n,B}^{\text{mb}}.$$

The bootstrap construction in (12) and (13) is conditionally Gaussian given the data. Thus its validity reduces to comparing the conditional Gaussian quadratic form with the oracle Gaussian reference. The next theorem gives a non-asymptotic bound for this comparison. Its proof uses a Gaussian comparison inequality for squared Euclidean norms and a nuclear-norm consistency bound for $\widehat{R}$.

Let $C_R > 0$ and $C_{\text{mb}} > 0$ be constants depending only on the constants in Assumptions 2.1 and 2.2. For $\eta_2 \in (0, 1)$, define

$$\rho_{\text{mb}}(\eta_2) = C_{\text{mb}} \kappa(R) \sqrt{\frac{d^3}{n_0 \eta_2}}, \quad n_0 = n_1 \wedge n_2. \quad (16)$$

Theorem 2.3 (Multiplier bootstrap approximation). *Suppose Assumptions 2.1, 2.2, and 2.3 hold. If*

$$C_R \sqrt{\frac{d^3}{n_0 \eta_2}} \leq \frac{1}{2} \Lambda_2,$$

then, with probability at least $1 - \eta_2$,

$$\sup_{r \geq 0} \left| \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \leq r \mid \mathcal{X} \right) - \mathbb{P}(T_Z \leq r) \right| \leq \rho_{\text{mb}}(\eta_2).$$

Combining Corollary 2.2 and Theorem 2.3 gives the validity of the bootstrap critical value (14) and the rejection rule (15).

Corollary 2.4 (Size validity). *Suppose $H_0 : \Sigma_1 = \Sigma_2$ holds. Suppose Assumptions 2.1, 2.2, and 2.3 hold. Assume further that the conditions of Corollary 2.2 and Theorem 2.3 are satisfied along the sequence, and that the error bounds in (11) and (16), together with the corresponding bootstrap failure probability $\rho_{\text{mb}}(\eta_2)$, converge to zero. Then the multiplier-bootstrap test has asymptotically correct size:*

$$\mathbb{P}_{H_0} \left(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}} \right) \rightarrow \alpha.$$

2.4 Power

Although the theoretical analysis is carried out after suppressing the superscript (A) , we restore it in this subsection to emphasize how the choice of transformation affects the alternative signal. Under $H_1 : \Sigma_1 \neq \Sigma_2$, define

$$\begin{aligned} \Delta_\Sigma^{(A)} &= \text{vech}\{A(\Sigma_1 - \Sigma_2)A^\top\}, \\ \theta_n^{(A)} &= \widetilde{D}^{-1/2} \Delta_\Sigma^{(A)} \in \mathbb{R}^d. \end{aligned} \tag{17}$$

Here $\theta_n^{(A)}$ in (17) is the standardized transformed covariance difference.

The following condition describes a primitive class of alternatives. Suppose there exist an index set $\mathcal{J}_n \subset \{1, \dots, d\}$, with $|\mathcal{J}_n| = m_n$, and a sequence $a_n > 0$ such that

$$\min_{k \in \mathcal{J}_n} \left| \theta_{n,k}^{(A)} \right| \geq a_n.$$

Thus at least m_n transformed covariance coordinates have standardized separation no smaller than a_n . The relevant signal accumulation is then

$$g_n = n_{\text{eff}} \frac{m_n}{d} a_n^2. \quad (18)$$

Theorem 2.5 (Power consistency). *Suppose $H_1 : \Sigma_1 \neq \Sigma_2$ holds. Suppose Assumptions 2.1 and 2.2 hold under the alternative. Assume that*

$$\frac{d}{n_0} \rightarrow 0, \quad n_0 = n_1 \wedge n_2,$$

and

$$\sqrt{\frac{n_{\text{eff}}}{d}} \left(\frac{p}{n_1} + \frac{p}{n_2} \right) = o(\sqrt{g_n}).$$

If

$$g_n = n_{\text{eff}} \frac{m_n}{d} a_n^2 \rightarrow \infty,$$

then

$$\mathbb{P}_{H_1} \left(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}} \right) \rightarrow 1.$$

Equivalently, the corresponding right-tail multiplier-bootstrap p-value converges to zero in probability.

The theorem shows that the test is consistent when the transformed covariance difference is sufficiently aggregated across coordinates, as measured by (18). In particular, dense alternatives with $m_n \asymp d$ require only $n_{\text{eff}} a_n^2 \rightarrow \infty$, whereas very sparse alternatives require a larger per-coordinate separation because the ℓ_2 -statistic averages over all $d = p(p+1)/2$ covariance coordinates. The role of A is therefore to increase the number or the magnitude of standardized transformed differences in (17) without creating excessive estimation error.

2.5 Unbalanced sample-size effect and simplified implementation

The simple quadratic structure of the proposed statistic gives a direct interpretation in highly unbalanced designs. The preceding power result shows that the signal is scaled by

the effective sample size n_{eff} in (3). When $n_1 \gg n_2$, n_{eff} is asymptotically equivalent to n_2 , and the contribution of the larger sample to the stochastic fluctuation becomes negligible. Consequently, the full two-sample statistic reduces to a one-sample covariance discrepancy statistic for the smaller sample, with the larger sample serving as a covariance reference.

Under H_0 , recall that

$$\widetilde{W} = \sqrt{n_{\text{eff}}}(\bar{H}_1 - \bar{H}_2).$$

When n_1 is much larger than n_2 , define the oracle one-sample reference vector

$$W_{2|1} = -\sqrt{n_2} \bar{H}_2, \tag{19}$$

and the associated oracle quadratic statistic

$$T_{2|1}^{\text{or}} = \frac{1}{d} \left\| D_2^{-1/2} W_{2|1} \right\|_2^2.$$

Proposition 2.2 (Unbalanced-sample reduction). Suppose $H_0 : \Sigma_1 = \Sigma_2$ holds and Assumptions 2.1 and 2.2 are satisfied. If

$$\varepsilon_n = \frac{n_2}{n_1} \rightarrow 0,$$

then

$$\frac{n_{\text{eff}}}{n_2} = 1 + O(\varepsilon_n), \quad \max_{1 \leq k \leq d} \left| \frac{\widetilde{D}_k}{D_{2,k}} - 1 \right| = O(\varepsilon_n),$$

and

$$\frac{1}{\sqrt{d}} \left\| \widetilde{D}^{-1/2} \widetilde{W} - D_2^{-1/2} W_{2|1} \right\|_2 = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

Consequently,

$$\widetilde{T}_n - T_{2|1}^{\text{or}} = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

Proposition 2.2 shows that, when $n_1/n_2 \rightarrow \infty$, the larger sample provides an asymptotically fixed covariance reference, while the stochastic fluctuation and the critical region are governed by the smaller-sample reference vector (19). The rate $O_{\mathbb{P}}\{(n_2/n_1)^{1/2}\}$ quantifies

how quickly this one-sample reduction becomes accurate. The symmetric conclusion holds when $n_2/n_1 \rightarrow \infty$, with the roles of the two samples interchanged.

The preceding result also suggests a simplified implementation in highly unbalanced settings. Suppose $n_1 \gg n_2$. Let $\widehat{\Sigma}_1$ be the covariance estimator from the larger sample and $\widehat{\Sigma}_2$ the covariance estimator from the smaller sample. Define

$$\widehat{W}_{2|1} = \sqrt{n_2} \text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_2).$$

For $k = k(a, b)$, $1 \leq a \leq b \leq p$, define the smaller-sample diagonal studentizer

$$\widehat{D}_{2,kk} = \frac{1}{n_2} \sum_{j=1}^{n_2} (\widehat{h}_{j,k}^{(2)})^2, \quad 1 \leq k \leq d,$$

and set

$$\widehat{T}_{2|1} = \frac{1}{d} \left\| \widehat{D}_2^{-1/2} \widehat{W}_{2|1} \right\|_2^2. \quad (20)$$

The statistic in (20) treats the larger sample as a covariance reference and uses only the smaller sample to estimate the stochastic fluctuation.

The ball geometry makes this reduction especially transparent. For a critical value c , the simplified acceptance region can be written as

$$\{\widehat{T}_{2|1} \leq c\} = \left\{ \left\| \widehat{D}_2^{-1/2} \text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_2) \right\|_2 \leq \sqrt{\frac{dc}{n_2}} \right\}. \quad (21)$$

Thus, in the studentized covariance-coordinate metric, the test accepts exactly when the larger-sample covariance estimator $\widehat{\Sigma}_1$ lies inside the one-sample confidence ball in (21), centered at the smaller-sample estimator $\widehat{\Sigma}_2$, with radius determined by the one-sample bootstrap critical value.

Corollary 2.6 (Feasible simplified statistic). *Suppose $H_0 : \Sigma_1 = \Sigma_2$ holds and Assumptions 2.1 and 2.2 are satisfied. If*

$$\varepsilon_n = \frac{n_2}{n_1} \rightarrow 0, \quad \frac{d}{n_2} \rightarrow 0,$$

then

$$\widehat{T}_{2|1} - T_{2|1}^{\text{or}} = O_{\mathbb{P}} \left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}} \right).$$

Moreover, if the full sample-to-oracle transfer satisfies

$$\widehat{T}_n - \widetilde{T}_n = o_{\mathbb{P}}(1),$$

then

$$\widehat{T}_n - \widehat{T}_{2|1} = o_{\mathbb{P}}(1).$$

More explicitly,

$$|\widehat{T}_n - \widehat{T}_{2|1}| \leq |\widehat{T}_n - \widetilde{T}_n| + O_{\mathbb{P}} \left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}} \right).$$

The simplified statistic can be calibrated by a one-sample multiplier bootstrap. Generate independent $N(0, 1)$ multipliers $e'_1, \dots, e'_{n_2}$ and define

$$\widehat{W}_{2|1}^{\text{mb}} = \frac{1}{\sqrt{n_2}} \sum_{j=1}^{n_2} e'_j \widehat{h}_j^{(2)},$$

with

$$\widehat{T}_{2|1}^{\text{mb}} = \frac{1}{d} \left\| \widehat{D}_2^{-1/2} \widehat{W}_{2|1}^{\text{mb}} \right\|_2^2.$$

The simplified bootstrap critical value is the conditional $(1 - \alpha)$ -quantile of $\widehat{T}_{2|1}^{\text{mb}}$. This simplified procedure should be viewed as an approximation to the full two-sample procedure in highly unbalanced regimes. The full statistic remains the default implementation, while $\widehat{T}_{2|1}$ provides a computationally and conceptually transparent approximation when n_2/n_1 is small.

3 Simulation Studies

The simulation studies are designed to evaluate the finite-sample behavior of the proposed procedure in regimes where the covariance-coordinate dimension $d = p(p + 1)/2$ is much

larger than the sample sizes. The numerical study is organized around two questions. Simulation 1 studies the proposed procedure itself: we compare the identity implementation $A = I_p$ with target transformations $A = G(\rho_{\text{tar}})$ for $\rho_{\text{tar}} \in \{0.3, 0.5, 0.7\}$. This part asks whether the target transformation improves finite-sample behavior and whether the procedure is sensitive to the exact target correlation. Simulation 2 then fixes the representative target $G(0.5)$ and compares it with existing covariance tests, both under the same null designs and under power alternatives. This separation is important: the first simulation justifies the transformation and the choice of a default target, while the second simulation evaluates the default method against external benchmarks.

The proposed statistic can be implemented in the original coordinate system, denoted by $A = I_p$, or after a transformation $A = G$. In applications, a transformation can be used to express the covariance difference in a geometry where dense covariance-coordinate signals are more visible. We implement this idea by mapping the empirical pooled covariance to a dense equicorrelation target. Specifically, let

$$\widehat{\Sigma}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} (X_{gi} - \bar{X}_g)(X_{gi} - \bar{X}_g)^\top, \quad g = 1, 2,$$

and

$$\widehat{\Sigma}_{\text{pool}} = \frac{n_1 \widehat{\Sigma}_1 + n_2 \widehat{\Sigma}_2}{n_1 + n_2}.$$

For a target correlation $\rho_{\text{tar}} \in \{0.3, 0.5, 0.7\}$, define the dense equicorrelation target

$$\Sigma_{\text{tar}}(\rho_{\text{tar}}) = (1 - \rho_{\text{tar}})I_p + \rho_{\text{tar}}\mathbf{1}_p\mathbf{1}_p^\top. \quad (22)$$

The numerical transformation used in the simulations is

$$G(\rho_{\text{tar}}) = \Sigma_{\text{tar}}(\rho_{\text{tar}})^{1/2} \left(\widehat{\Sigma}_{\text{pool}} + \lambda I_p \right)^{-1/2}, \quad \lambda = 10^{-6} \frac{\text{tr}(\widehat{\Sigma}_{\text{pool}})}{p}. \quad (23)$$

The ridge term in (23) is used only to compute the inverse square root stably. The theoretical results are stated for a deterministic transformation A ; $G(\rho_{\text{tar}})$ in (23) is the empirical

implementation used to study how target geometry affects finite-sample calibration and power.

The benchmark methods in Simulation 2 are a maximum-coordinate multiplier bootstrap test, the normal-reference method of Wang et al. (2024), and the two-sample covariance test of Li and Chen (2012). All simulations use independent samples with sample covariance matrices computed with denominator n , as in the proposed statistic, and the nominal level is $\alpha = 0.05$. The number of Monte Carlo repetitions R and multiplier-bootstrap draws B is stated in each simulation block. In the variance-heterogeneous power experiment, we also report a raw ℓ_2 bootstrap statistic, defined by removing the diagonal studentizer $\widehat{D}^{-1/2}$ from (6) while keeping the same quadratic aggregation and bootstrap calibration. This additional diagnostic separates the effect of coordinate studentization from the effect of target transformation.

We use the following covariance notation throughout the simulation section. For $\rho \in [0, 1)$, let

$$\Sigma_{\text{Eq}}(\rho) = (1 - \rho)I_p + \rho \mathbf{1}_p \mathbf{1}_p^\top$$

denote the equicorrelation covariance matrix, and let

$$\{\Sigma_{\text{Toe}}(\rho)\}_{ab} = \rho^{|a-b|}, \quad 1 \leq a, b \leq p,$$

denote the Toeplitz covariance matrix. In the tables, Eq and Toe abbreviate $\Sigma_{\text{Eq}}(\rho_0)$ and $\Sigma_{\text{Toe}}(\rho_0)$, respectively.

3.1 Simulation 1: target transformation and target sensitivity

We first examine calibration under the null hypothesis $\Sigma_1 = \Sigma_2$ for the proposed family of tests. The dimensions are $p \in \{100, 200\}$, and the sample sizes are $(n_1, n_2) \in \{(80, 100), (120, 150)\}$, with an additional small-sample check at $n_1 = n_2 = 60$. For each

dimension and sample-size pair, the common covariance is either $\Sigma_{\text{Eq}}(\rho_0)$ or $\Sigma_{\text{Toe}}(\rho_0)$, with $\rho_0 \in \{0.1, 0.5, 0.9\}$. The innovations are Gaussian, standardized centered χ_1^2 , and standardized Laplace. All formal size simulations use $R = 1000$ Monte Carlo repetitions and $B = 500$ multiplier draws for bootstrap methods. Tables 1 and 2 report the empirical rejection probabilities for the three target-transformed tests $G(0.3)$, $G(0.5)$, and $G(0.7)$, together with the identity implementation I_p . To keep the tables compact, each slash-separated triplet is ordered as $\rho_0 = 0.1, 0.5, 0.9$; Gau, Chi, and Lap denote Gaussian, standardized centered χ_1^2 , and standardized Laplace innovations, while Eq and Toe denote the two covariance families defined above.

Table 1: Empirical size of the proposed target-transformed and identity tests for $p = 100$. Each entry reports the $\rho_0 = 0.1/0.5/0.9$ triplet; abbreviations are defined in the text.

(n_1, n_2)	Dist.	Cov.	$G(.3)$	$G(.5)$	$G(.7)$	I_p
(60, 60)	Gau	Eq	0.046/0.049/0.057	0.050/0.065/0.062	0.051/0.062/0.069	0.001/0.054/0.060
		Toe	0.050/0.053/0.055	0.061/0.057/0.056	0.061/0.062/0.061	0.000/0.000/0.034
	Chi	Eq	0.061/0.047/0.050	0.062/0.046/0.054	0.064/0.043/0.045	0.005/0.045/0.049
		Toe	0.058/0.042/0.047	0.061/0.054/0.054	0.063/0.051/0.053	0.000/0.000/0.030
	Lap	Eq	0.048/0.053/0.040	0.062/0.060/0.044	0.058/0.060/0.044	0.003/0.062/0.047
		Toe	0.042/0.058/0.047	0.050/0.061/0.051	0.050/0.062/0.050	0.000/0.000/0.023
(80, 100)	Gau	Eq	0.046/0.054/0.044	0.050/0.054/0.053	0.047/0.052/0.049	0.005/0.050/0.054
		Toe	0.059/0.054/0.056	0.063/0.062/0.063	0.065/0.059/0.056	0.000/0.000/0.039
	Chi	Eq	0.058/0.049/0.059	0.058/0.045/0.062	0.062/0.054/0.061	0.010/0.048/0.061
		Toe	0.060/0.053/0.050	0.061/0.058/0.055	0.063/0.055/0.058	0.000/0.000/0.027
	Lap	Eq	0.062/0.056/0.054	0.066/0.060/0.057	0.063/0.057/0.056	0.008/0.054/0.058
		Toe	0.047/0.059/0.042	0.051/0.067/0.052	0.050/0.062/0.052	0.000/0.000/0.042
(120, 150)	Gau	Eq	0.058/0.060/0.051	0.062/0.066/0.054	0.057/0.070/0.053	0.011/0.073/0.049
		Toe	0.060/0.056/0.057	0.064/0.057/0.053	0.061/0.052/0.048	0.000/0.000/0.053
	Chi	Eq	0.064/0.040/0.050	0.057/0.040/0.044	0.064/0.039/0.048	0.016/0.040/0.040
		Toe	0.056/0.044/0.052	0.053/0.049/0.058	0.053/0.046/0.056	0.000/0.000/0.031
	Lap	Eq	0.052/0.047/0.052	0.056/0.051/0.054	0.065/0.049/0.052	0.008/0.052/0.054
		Toe	0.042/0.058/0.053	0.044/0.062/0.051	0.044/0.061/0.053	0.000/0.000/0.036

Although the ambient dimensions $p = 100$ and $p = 200$ appear moderate, the testing problem is high-dimensional because the statistic aggregates over the covariance-coordinate dimension $d = p(p+1)/2$. Thus $d = 5050$ when $p = 100$ and $d = 20100$ when $p = 200$, both far exceeding the sample sizes used in the simulation, including the small-sample setting $n_1 = n_2 = 60$ that satisfies the prespecified calibration check and is therefore included in the main table. This regime is precisely where direct estimation of the full covariance matrix of covariance entries is unstable, and where the proposed diagonal studentization

and multiplier-bootstrap calibration are intended to be useful.

Table 2: Empirical size of the proposed target-transformed and identity tests for $p = 200$. Each entry reports the $\rho_0 = 0.1/0.5/0.9$ triplet; abbreviations are defined in the text.

(n_1, n_2)	Dist.	Cov.	$G(.3)$	$G(.5)$	$G(.7)$	I_p
(60, 60)	Gau	Eq	0.050/0.062/0.042	0.051/0.068/0.044	0.056/0.066/0.041	0.001/0.063/0.043
		Toe	0.052/0.042/0.054	0.056/0.046/0.056	0.061/0.046/0.057	0.000/0.000/0.011
	Chi	Eq	0.042/0.056/0.047	0.049/0.056/0.046	0.045/0.054/0.044	0.008/0.059/0.044
		Toe	0.045/0.059/0.043	0.045/0.059/0.044	0.049/0.055/0.047	0.000/0.000/0.005
	Lap	Eq	0.053/0.046/0.051	0.056/0.046/0.052	0.055/0.046/0.051	0.001/0.045/0.049
		Toe	0.054/0.060/0.059	0.052/0.056/0.058	0.056/0.060/0.054	0.000/0.000/0.007
(80, 100)	Gau	Eq	0.059/0.053/0.057	0.060/0.050/0.061	0.060/0.053/0.056	0.004/0.053/0.058
		Toe	0.053/0.050/0.052	0.053/0.049/0.054	0.056/0.051/0.056	0.000/0.000/0.015
	Chi	Eq	0.050/0.069/0.052	0.053/0.064/0.053	0.048/0.061/0.057	0.006/0.073/0.053
		Toe	0.044/0.054/0.056	0.048/0.063/0.057	0.048/0.056/0.056	0.000/0.000/0.011
	Lap	Eq	0.061/0.061/0.051	0.067/0.062/0.055	0.066/0.072/0.053	0.005/0.065/0.058
		Toe	0.047/0.049/0.053	0.052/0.048/0.049	0.056/0.051/0.057	0.000/0.000/0.016
(120, 150)	Gau	Eq	0.049/0.054/0.058	0.053/0.053/0.060	0.053/0.058/0.056	0.005/0.051/0.061
		Toe	0.066/0.051/0.046	0.070/0.057/0.056	0.066/0.057/0.054	0.000/0.000/0.024
	Chi	Eq	0.049/0.054/0.043	0.053/0.052/0.047	0.046/0.049/0.050	0.013/0.046/0.044
		Toe	0.070/0.066/0.061	0.075/0.068/0.061	0.070/0.066/0.061	0.000/0.000/0.020
	Lap	Eq	0.044/0.050/0.039	0.046/0.051/0.042	0.052/0.050/0.042	0.006/0.048/0.041
		Toe	0.050/0.048/0.049	0.054/0.053/0.052	0.051/0.052/0.052	0.000/0.000/0.025

The main conclusion from the size study is that the target transformation stabilizes the proposed test. Across the 108 main null settings, the mean empirical sizes of $G(0.3)$, $G(0.5)$, and $G(0.7)$ are 0.052, 0.055, and 0.055, respectively, with all setting-level rejection rates lying between 0.039 and 0.075. Thus the transformed tests remain close to the nominal level over changes in p , sample size, covariance family, dependence strength, and innovation distribution. The three target choices also produce very similar calibration patterns, indicating that the transfer to a dense target geometry is not especially sensitive to the precise value of $\rho_{\text{tar}} \in \{0.3, 0.5, 0.7\}$. In contrast, the identity implementation has mean empirical size 0.023 and is often conservative, especially in weakly dependent Toeplitz designs. This comparison shows that the gain from $G(\rho_{\text{tar}})$ is not a fragile tuning effect; it comes from re-expressing the covariance difference in a geometry that is better aligned with dense ℓ_2 aggregation.

We next examine power within the proposed family under dense equicorrelation alternatives. The power studies use $p = 200$, $n_1 = 120$, $n_2 = 150$, Gaussian samples, nominal level 0.05, $R = 1000$ Monte Carlo repetitions, and $B = 500$ multiplier draws for bootstrap

methods. The first experiment sets $\Sigma_1 = \Sigma_{\text{Eq}}(0)$ and $\Sigma_2 = \Sigma_{\text{Eq}}(\rho_1)$, where ρ_1 increases from zero. The second sets $\Sigma_1 = \Sigma_{\text{Eq}}(0.5)$ and $\Sigma_2 = \Sigma_{\text{Eq}}(\rho_1)$, with $\rho_1 \in \{0.1, \dots, 0.9\}$. In both experiments, $G(0.3)$, $G(0.5)$, and $G(0.7)$ have nearly overlapping power curves. The maximum setting-level difference among the three target-transformed rejection rates is 0.011 in the $\rho_0 = 0$ experiment and 0.021 in the $\rho_0 = 0.5$ experiment. Relative to the identity implementation, the transformation is especially beneficial near the identity baseline: when $\rho_0 = 0$, I_p has essentially no rejection before $\rho_1 = 0.03$, whereas $G(0.5)$ already has rejection rates 0.387 and 0.922 at $\rho_1 = 0.002$ and $\rho_1 = 0.005$, respectively. We therefore use $G(0.5)$ as the default target in the remaining comparisons: it has stable size, strong power, and is representative of the target-transformed family rather than a specially tuned choice.

3.2 Simulation 2: default target versus benchmark tests

We now fix the proposed method at $G(0.5)$ and compare it with the existing tests. Under the same null grid used in Simulation 1, the max-bootstrap and Wang-NR tests are often conservative, with mean empirical sizes 0.042 and 0.025, respectively, while Li-Chen is liberal in several correlated non-Gaussian settings, with mean empirical size 0.105 and maximum empirical size 0.269. These size results provide the calibration context for the power comparisons below: rejection rates away from the null are most meaningful when the corresponding method is also well calibrated near the null.

The first benchmark power comparison uses the two equicorrelation designs described above. The results are reported graphically because the scientific question is the shape of the power curves as the alternative strengthens, rather than isolated rejection rates at a single point.

Figure 1 shows that $G(0.5)$ is highly sensitive to dense covariance alternatives while preserving the calibration pattern observed under the null. When $\rho_0 = 0$, even very small positive

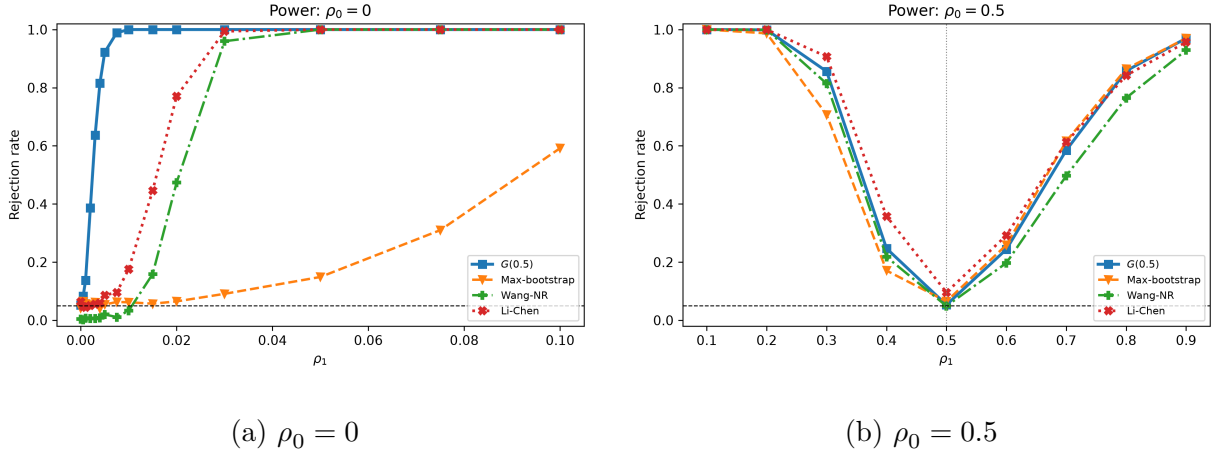


Figure 1: Empirical rejection rates for dense equicorrelation alternatives. Panel (a) compares $\Sigma_{\text{Eq}}(0)$ with $\Sigma_{\text{Eq}}(\rho_1)$. Panel (b) compares $\Sigma_{\text{Eq}}(0.5)$ with $\Sigma_{\text{Eq}}(\rho_1)$. The proposed method uses the default target $G(0.5)$ and is compared with max-bootstrap, Wang-NR, and Li-Chen. Results are based on $R = 1000$ Monte Carlo repetitions and $B = 500$ multiplier draws for bootstrap methods.

equicorrelations produce a covariance-coordinate signal spread over many entries: $G(0.5)$ has power 0.387 at $\rho_1 = 0.002$, 0.922 at $\rho_1 = 0.005$, and reaches 1.000 by $\rho_1 = 0.010$. Over the same range, max-bootstrap remains near the nominal level, and Li-Chen increases more slowly. When $\rho_0 = 0.5$, the $G(0.5)$ curve is lowest at the null point $\rho_1 = 0.5$, with rejection rate 0.053, and increases as ρ_1 moves away from the correlated baseline. The benchmark methods also gain power for large departures, but the default target-transformed test is especially competitive for weak and moderate dense alternatives while maintaining the size behavior established in Simulation 1.

The second benchmark power comparison is designed to isolate the statistical role of studentization. Both groups have independent coordinates under the null. The first 80 coordinates form a low-variance block with marginal variance one, and the remaining 120 coordinates are high-variance nuisance coordinates with marginal variance 25. Under the alternative, only the low-variance block changes: its within-block correlation is increased from zero

to ρ_{block} , while all other correlations and all marginal variances remain unchanged. Thus $\rho_{\text{block}} = 0$ is the null case. This construction creates a dense covariance signal in coordinates whose raw Frobenius contribution can be masked by high-variance nuisance coordinates.

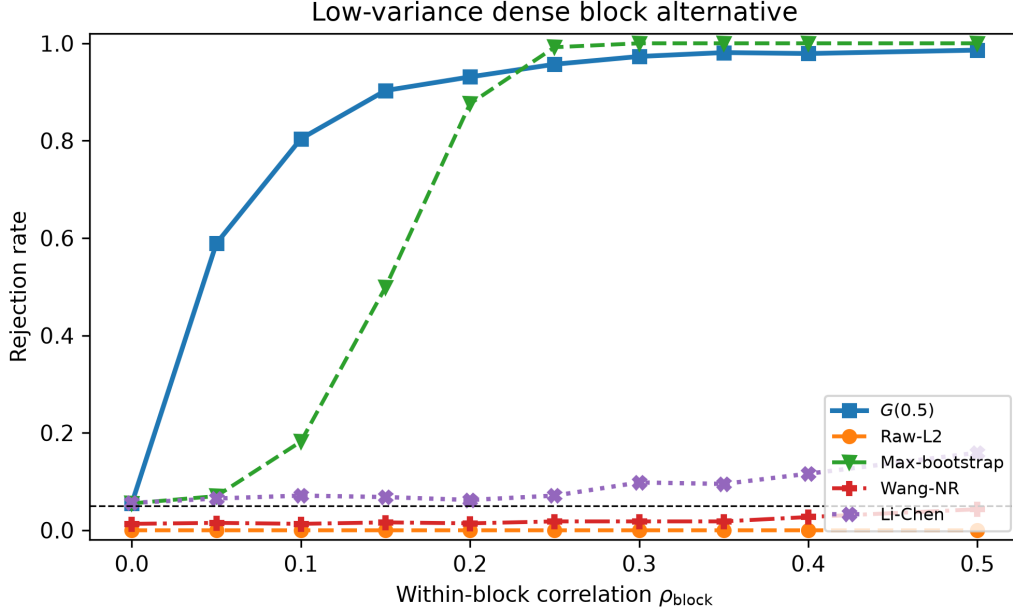


Figure 2: Empirical rejection rates for the variance-heterogeneous low-variance block alternative. The first 80 coordinates carry the dense block-correlation signal and have marginal variance one; the remaining 120 coordinates have marginal variance 25 and act as high-variance nuisance coordinates. Results are based on $R = 1000$ Monte Carlo repetitions and $B = 500$ multiplier draws for bootstrap methods.

Figure 2 demonstrates that studentization is not merely a technical normalization. The raw ℓ_2 statistic, which uses the unstudentized quadratic norm of the covariance-coordinate difference, has essentially no power in this design because the high-variance nuisance coordinates dominate the raw scale. By contrast, $G(0.5)$ is close to nominal size at $\rho_{\text{block}} = 0$ and becomes powerful for weak within-block correlations: its rejection rate is 0.589 at $\rho_{\text{block}} = 0.05$, 0.804 at $\rho_{\text{block}} = 0.10$, and 0.903 at $\rho_{\text{block}} = 0.15$. The max-bootstrap test becomes powerful only at larger block correlations, while Wang-NR and Li-Chen have

little power in this particular variance-heterogeneous design. This comparison shows that the default target-transformed test is not only calibrated in the main size study but also effective against dense covariance signals that can be hidden by heterogeneous marginal scales.

4 Real Data Application

4.1 TCGA-BRCA covariance analysis

We apply the proposed covariance-testing framework to TCGA-BRCA RNA-seq data, comparing Basal-like and Luminal A breast-cancer subtypes. This comparison is biologically well motivated because TCGA molecular profiling shows that these breast-cancer subtypes have distinct molecular and transcriptional profiles ([The Cancer Genome Atlas Network, 2012](#)). In contrast to conventional analyses that focus primarily on marginal expression differences, our objective is to examine whether a biologically coherent transcriptional program exhibits subtype-specific covariance organization.

Expression and clinical annotation files were obtained from the UCSC Xena TCGA-BRCA cohort ([Goldman et al., 2020](#)), using the `HiSeqV2.gz` expression matrix and `BRCA_clinicalMatrix` clinical annotation file. We retained primary tumor samples with PAM50 RNA-seq subtype labels Basal-like or Luminal A. The resulting groups contain $n_1 = 141$ Basal-like tumors and $n_2 = 421$ Luminal A tumors. Expression values are analyzed on the Xena log-transformed RNA-seq scale. Genes are matched by symbol to the MSigDB Hallmark file `h.all.v2025.1.Hs.symbols.gmt` ([Liberzon et al., 2015](#)); duplicate gene symbols are averaged, and genes with negligible variance are removed.

The application focuses on a Hallmark cell-cycle core panel formed by the union of the complete Hallmark E2F targets and G2M checkpoint gene sets. These two Hallmark com-

ponents represent closely related transcriptional programs governing cell-cycle entry, DNA replication, and mitotic progression, and are therefore directly relevant to the proliferative phenotype that distinguishes Basal-like tumors from Luminal A tumors. After matching to the expression matrix and applying the same low-variance filtering described above, the panel contains $p = 304$ genes. Since $p \geq \min(n_1, n_2)$, this application remains in the high-dimensional regime targeted by the proposed methodology.

We compare the proposed bootstrap statistic with $A = I_p$, the target-transformed implementation with $A = G$, max-bootstrap, Wang-NR, and Li-Chen. Bootstrap methods use $B = 2000$ multiplier draws. The table reports program-level p -values for this focused biological application; these values are interpreted as evidence for covariance reorganization within the defined cell-cycle program rather than as a genome-wide discovery claim.

Table 3: TCGA-BRCA Basal-like versus Luminal A covariance comparison for the Hallmark cell-cycle core panel. The table reports the observed test statistic, program-level p -value, and rejection decision. For bootstrap methods, < 0.0005 means that no bootstrap draw exceeded the observed statistic among $B = 2000$ draws.

method	statistic	p -value	reject
$A = I_p$	4.239	< 0.0005	yes
$A = G$	7.232	< 0.0005	yes
Max	11.980	< 0.0005	yes
Wang-NR	1777.476	5.1×10^{-13}	yes
Li-Chen	27.004	6.6×10^{-161}	yes

Table 3 shows strong evidence of covariance reorganization in the Hallmark cell-cycle core panel. Both proposed implementations reject the null hypothesis: the untransformed statistic with $A = I_p$ and the target-transformed implementation with $A = G$ each yield bootstrap $p < 0.0005$. The same conclusion is supported by the benchmark procedures, with

max-bootstrap giving $p < 0.0005$, Wang-NR giving $p = 5.1 \times 10^{-13}$, and Li-Chen giving $p = 6.6 \times 10^{-161}$. Thus, the empirical finding is not an artifact of a particular calibration scheme. The agreement between $A = I_p$ and $A = G$ is also consistent with the dense program-level covariance signal emphasized in the simulation studies.

The selected panel has a direct biological interpretation. It contains 187 matched E2F-target genes and 183 matched G2M-checkpoint genes, with 304 unique genes after accounting for overlap between the two Hallmark components. These components capture coordinated transcriptional activity associated with DNA replication, cell-cycle progression, and mitotic control, all of which are central to the proliferative biology of Basal-like breast tumors relative to Luminal A tumors. As a descriptive check on the source of the signal, the mean absolute difference between the two subtype-specific sample correlations is 0.145 within the E2F-target component, 0.137 within the G2M-checkpoint component, and 0.141 across the two components. The covariance-level rejection therefore has a natural program-level interpretation: the two subtypes differ not only in the expression levels of proliferation-related genes, but also in the pattern of gene-gene coordination within this cell-cycle program.

4.2 Real-data-calibrated simulation

As an additional calibration check tied directly to this application, we conducted a real-data-calibrated Gaussian simulation. The sample sizes and dimension were fixed at the observed values $n_1 = 141$, $n_2 = 421$, and $p = 304$. For the size experiment, both samples were generated from a common covariance matrix equal to the pooled empirical covariance from the selected panel. For the power experiment, the two samples were generated from the subtype-specific empirical covariance matrices estimated from the Basal-like and Luminal A groups, respectively. This design preserves the scale and correlation geometry of the

application while making the null and alternative data-generating mechanisms explicit.

Table 4: Real-data-calibrated Gaussian simulation for the TCGA-BRCA cell-cycle core panel. Size uses the pooled empirical covariance as the common covariance matrix; power uses the two subtype-specific empirical covariance matrices. Results are based on $R = 1000$ Monte Carlo repetitions and $B = 2000$ bootstrap draws for bootstrap methods.

method	size	power
$A = I_p$	0.071	1.000
$A = G$	0.065	1.000
Max-bootstrap	0.134	1.000
Wang-NR	0.062	1.000
Li-Chen	0.112	1.000

Table 4 supports the interpretation of the application. Under the pooled-covariance null, the two proposed bootstrap tests have empirical rejection rates 0.071 for $A = I_p$ and 0.065 for $A = G$, close to the nominal level. Wang-NR is similar, whereas max-bootstrap and Li-Chen are more liberal in this high-dimensional empirical-covariance design. Under the subtype-specific covariance alternative, all methods reject in every repetition, indicating that the covariance contrast estimated from the real data is strong under Gaussian resampling.

5 Conclusion and Discussion

This paper develops a studentized ℓ_2 approach to two-sample covariance testing. The method compares covariance matrices through standardized upper-triangular covariance-coordinate differences and aggregates them by a squared Euclidean norm. This construction

gives a transparent scale-normalized global statistic, avoids estimating the full covariance matrix of covariance entries, and leads naturally to centered-ball Gaussian approximation. We prove oracle Gaussian approximation, feasible transfer from oracle to sample-centered studentized statistics, multiplier-bootstrap validity, and power consistency under accumulated standardized covariance signal. The same quadratic structure also yields a direct simplification in highly unbalanced designs, where the larger sample acts as a covariance reference and the smaller sample governs the stochastic fluctuation.

The simulations support the intended use of the procedure. The proposed bootstrap tests have stable empirical size across the null settings considered, and the target-transformed implementation $A = G$ is particularly effective for dense equicorrelation alternatives when compared among methods with acceptable size behavior. The untransformed statistic $A = I_p$ remains a useful baseline, while max-type procedures are less sensitive to diffuse covariance changes and some quadratic-form benchmarks can exhibit size distortion in correlated designs. In the TCGA-BRCA application, both proposed implementations detect a clear covariance reorganization in the Hallmark cell-cycle core panel between Basal-like and Luminal A tumors, and the real-data-calibrated simulation supports the interpretation that the observed covariance contrast is strong rather than a calibration artifact.

The theory is deliberately explicit, but the resulting high-dimensional sufficient conditions are conservative. In particular, the proof uses uniform covariance-coordinate controls and Gaussian comparison bounds that are not expected to be sharp for all covariance geometries. The simulation results suggest that the bootstrap statistic can perform well in regimes beyond those directly captured by the current growth conditions. A natural next step is to replace worst-case dimension dependence with sharper effective-rank, spectral-decay, or geometry-adaptive quantities, which may better reflect the behavior observed in finite samples.

Another important direction is the choice of the transformation A . The power analysis shows that A affects the standardized signal by changing how the covariance difference is represented before aggregation. This raises a broader question: whether one can characterize, estimate, or approximate transformations that are globally more powerful for classes of alternatives while retaining reliable bootstrap calibration. The dense equicorrelation target used here is one practical default, but it is unlikely to be optimal in all scientific settings. Developing principled rules for selecting A , and understanding the tradeoff between signal enhancement and estimation stability, would make the framework more adaptive and potentially more powerful.

6 Disclosure statement

A disclosure statement should be completed before submission.

7 Data Availability Statement

The simulation and real-data analysis code are available at <https://github.com/shuhaozheng1/covtestsim>. The TCGA-BRCA expression and clinical annotations used in the application are publicly available from UCSC Xena, and the Hallmark gene-set definitions are publicly available from MSigDB. The real-data pipeline in the code repository downloads these public resources and reproduces the processed analysis files.

SUPPLEMENTARY MATERIAL

Technical supplement: Auxiliary lemmas and proofs are included below in the appendix of this combined manuscript.

Code repository: The simulation and real-data analysis code are organized as a

standalone code repository, with source code, command-line runners, tests, and a README for reproducing the empirical results, at <https://github.com/shuhaozhen1/covtestsim>.

Supplementary Material

Simple Studentized ℓ_2 Statistics and Gaussian Bootstrap for Two-Sample Covariance Testing

Haozhen Shu and Tianming Zhu

This appendix contains the auxiliary lemmas and proofs for the main manuscript. Equation, theorem, proposition, corollary, and assumption numbers refer to the main manuscript unless otherwise stated.

A Auxiliary Lemmas and Proofs

A.1 Auxiliary lemmas

The proofs in this supplement rely on four external results. We collect them here for ease of reference. The constants appearing in these lemmas are universal (i.e., independent of n_1, n_2, d) unless stated otherwise.

Lemma A.1 (Gaussian approximation on Euclidean balls; [Fang and Koike, 2021](#), Theorem 3.2). *Let $\zeta_1, \dots, \zeta_N$ be independent centered random vectors in $\mathbb{R}^d$ with finite fourth moments, and set $W_\zeta = \sum_{\ell=1}^N \zeta_\ell$ and $\Sigma_W = \text{Var}(W_\zeta)$. Let $Z \sim N(0, \Sigma)$ and assume $\Lambda_2(\Sigma) > 0$. Then*

$$\sup_{r \geq 0} |\mathbb{P}(\|W_\zeta\|_2 \leq r) - \mathbb{P}(\|Z\|_2 \leq r)| \leq K \{ \kappa(\Sigma)^{3/4} \delta_1(W_\zeta)^{1/4} + \kappa(\Sigma)^{2/3} \delta_2(W_\zeta)^{1/3} \\ + \kappa(\Sigma)^{2/3} \delta_0(\Sigma_W, \Sigma)^{1/3} + \kappa(\Sigma)^{1/2} \delta'_0(\Sigma_W, \Sigma)^{1/2} \},$$

where $K_0 > 0$ is a universal absolute constant and

$$\delta_0(\Sigma_W, \Sigma) = \sqrt{(\text{tr}(\Sigma_W) + \text{tr}(\Sigma))(\|\Sigma_W\|_{\text{op}} + \|\Sigma\|_{\text{op}})} \|\Sigma - \Sigma_W\|_{\text{HS}},$$

$$\begin{aligned}\delta'_0(\Sigma_W, \Sigma) &= \sum_{j=1}^d |\Sigma_{jj} - \Sigma_{W,jj}|, \\ \delta_1(W_\zeta) &= \|\Sigma_W\|_{\text{HS}} \sum_{\ell=1}^N \mathbb{E} \|\zeta_\ell\|_2^4 + \|\Sigma_W\|_{\text{op}}^{3/2} \sum_{\ell=1}^N \mathbb{E} \|\zeta_\ell\|_2^3, \\ \delta_2(W_\zeta) &= \|\Sigma_W\|_{\text{op}}^{1/2} \sum_{\ell=1}^N \mathbb{E} \|\zeta_\ell\|_2^3 + \sum_{\ell=1}^N \mathbb{E} \|\zeta_\ell\|_2^4.\end{aligned}$$

Lemma A.2 (Anti-concentration for the squared norm of a Gaussian vector; [Fang and Koike, 2021](#), Lemma 5.10, originating from [Götze et al., 2019](#), Theorem 2.7). *Let $Z \sim N(0, \Sigma)$ with $\Lambda_2(\Sigma) > 0$. Then for every $\mu \in \mathbb{R}^d$ and every $\varepsilon > 0$,*

$$\sup_{a \geq 0} \mathbb{P}(a \leq \|Z + \mu\|_2^2 \leq a + \varepsilon) \leq K \kappa(\Sigma) \varepsilon,$$

where $K_0 > 0$ is a universal absolute constant.

Lemma A.3 (Gaussian comparison for squared Euclidean norms). *Let R_1, R_2 be $d \times d$ symmetric positive semidefinite matrices, and let $Z_m \sim N(0, R_m)$ for $m = 1, 2$. For each m , let*

$$\lambda_1(R_m) \geq \dots \geq \lambda_d(R_m) \geq 0$$

be the eigenvalues of R_m , and define

$$\Lambda_s(R_m) = \left(\sum_{\ell=s}^d \lambda_\ell(R_m)^2 \right)^{1/2}, \quad s = 1, 2.$$

Assume

$$\Lambda_2(R_1) > 0, \quad \Lambda_2(R_2) > 0,$$

and set

$$\kappa_*(R_m) = \{\Lambda_1(R_m) \Lambda_2(R_m)\}^{-1/2}.$$

Then there exists a universal constant $K > 0$ such that

$$\sup_{r \geq 0} |\mathbb{P}(\|Z_1\|_2^2 \leq r) - \mathbb{P}(\|Z_2\|_2^2 \leq r)| \leq K \{\kappa_*(R_1) + \kappa_*(R_2)\} \|R_1 - R_2\|_{S_1},$$

where $\|\cdot\|_{S_1}$ denotes the Schatten-1, or nuclear, norm.

Proof. We apply Theorem 2.1 and Corollary 2.3 of [Götze et al. \(2019\)](#). Their Gaussian comparison result gives, for two centered Gaussian elements with covariance operators R_1 and R_2 ,

$$\sup_{x \geq 0} |\mathbb{P}(\|Z_1\|_2 \leq x) - \mathbb{P}(\|Z_2\|_2 \leq x)| \leq K\{\kappa_G(R_1) + \kappa_G(R_2)\}\|R_1 - R_2\|_{S_1},$$

where $\kappa_G(\cdot)$ denotes the geometry factor used in [Götze et al. \(2019\)](#). Corollary 2.3 of that paper is precisely the nuclear-norm version of the comparison bound.

Their $\kappa_G(R)$ is equivalent, up to universal constants, to

$$\{\Lambda_1(R)\Lambda_2(R)\}^{-1/2}.$$

Therefore, after absorbing this equivalence into the universal constant K , we obtain

$$\sup_{x \geq 0} |\mathbb{P}(\|Z_1\|_2 \leq x) - \mathbb{P}(\|Z_2\|_2 \leq x)| \leq K\{\kappa_*(R_1) + \kappa_*(R_2)\}\|R_1 - R_2\|_{S_1}.$$

Finally, the map $r \mapsto \sqrt{r}$ is a bijection from $[0, \infty)$ to $[0, \infty)$. Hence

$$\sup_{r \geq 0} |\mathbb{P}(\|Z_1\|_2^2 \leq r) - \mathbb{P}(\|Z_2\|_2^2 \leq r)| = \sup_{x \geq 0} |\mathbb{P}(\|Z_1\|_2 \leq x) - \mathbb{P}(\|Z_2\|_2 \leq x)|.$$

This proves the claim. □

Lemma A.4 (Hoffman–Wielandt inequality). *Let A, B be $d \times d$ real symmetric matrices with eigenvalues $\lambda_1(A) \geq \dots \geq \lambda_d(A)$ and $\lambda_1(B) \geq \dots \geq \lambda_d(B)$. Then*

$$\sum_{\ell=1}^d (\lambda_\ell(A) - \lambda_\ell(B))^2 \leq \|A - B\|_F^2.$$

Lemma A.5 (Euclidean moment consequences of coordinate moments). *Suppose Assumption 2.1 holds. Then there exists a constant $C < \infty$, depending only on K_8 , such that*

$$\text{tr}\{\text{Cov}(U_i)\} + \text{tr}\{\text{Cov}(V_j)\} \leq Cp,$$

$$\mathbb{E}\|U_i\|_2^4 + \mathbb{E}\|V_j\|_2^4 \leq Cp^2, \quad \mathbb{E}\|U_i\|_2^8 + \mathbb{E}\|V_j\|_2^8 \leq Cp^4.$$

Moreover, for independent centered observations satisfying the same coordinate moment bound,

$$\mathbb{E}\|\bar{U}\|_2^4 \leq C \left(\frac{p}{n_1}\right)^2, \quad \mathbb{E}\|\bar{U}\|_2^8 \leq C \left(\frac{p}{n_1}\right)^4,$$

and the analogous bounds hold for $\bar{V}$. Finally,

$$\mathbb{E} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|U_i - \bar{U}\|_2^2 \right)^4 \leq Cp^4,$$

with the same conclusion for the second sample.

Proof. We prove the statements for U_i ; the argument for V_j is identical. By Lyapunov's inequality, $\mathbb{E}U_{ia}^2 \leq K_8^{1/4}$ and $\mathbb{E}U_{ia}^4 \leq K_8^{1/2}$ uniformly in a . Hence

$$\text{tr}\{\text{Cov}(U_i)\} \leq \sum_{a=1}^p \mathbb{E}U_{ia}^2 \leq Cp.$$

For the fourth moment, with $z_a = U_{ia}^2$,

$$\|U_i\|_2^4 = \left(\sum_{a=1}^p z_a \right)^2 \leq p \sum_{a=1}^p z_a^2 = p \sum_{a=1}^p U_{ia}^4,$$

so $\mathbb{E}\|U_i\|_2^4 \leq Cp^2$. Similarly,

$$\|U_i\|_2^8 = \left(\sum_{a=1}^p z_a \right)^4 \leq p^3 \sum_{a=1}^p z_a^4 = p^3 \sum_{a=1}^p U_{ia}^8,$$

and therefore $\mathbb{E}\|U_i\|_2^8 \leq Cp^4$.

For the sample mean, the standard scalar Rosenthal inequality gives

$$\mathbb{E}|\bar{U}_a|^4 \leq Cn_1^{-2}, \quad \mathbb{E}|\bar{U}_a|^8 \leq Cn_1^{-4}$$

uniformly in a . Applying the preceding sum inequalities to $\bar{U} = (\bar{U}_1, \dots, \bar{U}_p)^\top$ yields

$$\mathbb{E}\|\bar{U}\|_2^4 \leq C \left(\frac{p}{n_1}\right)^2, \quad \mathbb{E}\|\bar{U}\|_2^8 \leq C \left(\frac{p}{n_1}\right)^4.$$

Finally,

$$\frac{1}{n_1} \sum_{i=1}^{n_1} \|U_i - \bar{U}\|_2^2 \leq \frac{2}{n_1} \sum_{i=1}^{n_1} \|U_i\|_2^2 + 2\|\bar{U}\|_2^2.$$

Using $(x+y)^4 \leq 8(x^4+y^4)$, Jensen's inequality, and the eighth-moment bound just proved,

$$\mathbb{E} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|U_i - \bar{U}\|_2^2 \right)^4 \leq C \mathbb{E} \|U_i\|_2^8 + C \mathbb{E} \|\bar{U}\|_2^8 \leq Cp^4.$$

This proves the lemma. $\square$

A.2 Proofs

Proof of Theorem 2.1. Throughout the proof, we work in the transformed model and suppress the superscript (A) . Under $H_0 : \Sigma_1 = \Sigma_2$, the transformed covariance matrices are equal.

For $1 \leq a \leq b \leq p$, let $k = k(a, b)$ denote the corresponding coordinate of the half-vectorization. Define

$$H_i^{(1)} = \text{vech}\{(X_i - \mu_1)(X_i - \mu_1)^\top - \Sigma_1\},$$

and

$$H_j^{(2)} = \text{vech}\{(Y_j - \mu_2)(Y_j - \mu_2)^\top - \Sigma_2\}.$$

Equivalently, in the notation of the primitive assumptions,

$$H_{i,k}^{(1)} = U_{ia}U_{ib} - \mathbb{E}(U_{ia}U_{ib}), \quad H_{j,k}^{(2)} = V_{ja}V_{jb} - \mathbb{E}(V_{ja}V_{jb}).$$

We first record the standardized moment bounds for these covariance-coordinate summands.

By Assumption 2.1, for every $1 \leq a \leq b \leq p$,

$$\mathbb{E}|U_{ia}U_{ib}|^4 \leq \{\mathbb{E}|U_{ia}|^8 \mathbb{E}|U_{ib}|^8\}^{1/2} \leq K_8,$$

and similarly for V_j . Hence,

$$\mathbb{E} |U_{ia}U_{ib} - \mathbb{E}(U_{ia}U_{ib})|^4 \leq 16K_8.$$

By Lyapunov's inequality,

$$\mathbb{E} |U_{ia}U_{ib} - \mathbb{E}(U_{ia}U_{ib})|^3 \leq (16K_8)^{3/4}.$$

The same bounds hold for the second sample. Together with Assumption 2.2, this implies that, for $q = 3, 4$,

$$\max_{m=1,2} \max_{1 \leq k \leq d} \frac{\mathbb{E}|H_{1,k}^{(m)}|^q}{\text{Var}(H_{1,k}^{(m)})^{q/2}} \leq M_q,$$

where $M_q < \infty$ depends only on K_8 and c_H .

Next, under H_0 ,

$$\widetilde{W} = \sqrt{n_{\text{eff}}} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} H_i^{(1)} - \frac{1}{n_2} \sum_{j=1}^{n_2} H_j^{(2)} \right\}.$$

For each coordinate k ,

$$\widetilde{D}_{kk} = \text{Var}(\widetilde{W}_k) = n_{\text{eff}} \left\{ \frac{\text{Var}(H_{1,k}^{(1)})}{n_1} + \frac{\text{Var}(H_{1,k}^{(2)})}{n_2} \right\}.$$

By Assumption 2.2,

$$\text{Var}(H_{1,k}^{(1)}) \geq c_H, \quad \text{Var}(H_{1,k}^{(2)}) \geq c_H.$$

Therefore,

$$\widetilde{D}_{kk} \geq n_{\text{eff}} c_H \left(\frac{1}{n_1} + \frac{1}{n_2} \right) = c_H,$$

because

$$n_{\text{eff}} = \frac{n_1 n_2}{n_1 + n_2}.$$

Thus $\widetilde{D}^{-1/2}$ is well defined.

Define independent centered random vectors

$$\xi_i = \frac{\sqrt{n_{\text{eff}}}}{n_1} \widetilde{D}^{-1/2} H_i^{(1)}, \quad 1 \leq i \leq n_1,$$

and

$$\xi_{n_1+j} = -\frac{\sqrt{n_{\text{eff}}}}{n_2} \widetilde{D}^{-1/2} H_j^{(2)}, \quad 1 \leq j \leq n_2.$$

Let

$$S_n = \sum_{\ell=1}^{n_1+n_2} \xi_\ell.$$

Then

$$S_n = \widetilde{D}^{-1/2} \widetilde{W}. \tag{24}$$

Therefore,

$$\tilde{T}_n = \frac{1}{d} \|S_n\|_2^2. \quad (25)$$

Moreover, by independence of the ξ_ℓ 's,

$$\text{Cov}(S_n) = \sum_{\ell=1}^{n_1+n_2} \text{Cov}(\xi_\ell) = \sum_{i=1}^{n_1} \frac{n_{\text{eff}}}{n_1^2} \tilde{D}^{-1/2} \Gamma_1 \tilde{D}^{-1/2} + \sum_{j=1}^{n_2} \frac{n_{\text{eff}}}{n_2^2} \tilde{D}^{-1/2} \Gamma_2 \tilde{D}^{-1/2},$$

where $\Gamma_1 = \text{Cov}(H_i^{(1)})$ and $\Gamma_2 = \text{Cov}(H_j^{(2)})$. Simplifying,

$$\text{Cov}(S_n) = \tilde{D}^{-1/2} \left\{ n_{\text{eff}} \left(\frac{\Gamma_1}{n_1} + \frac{\Gamma_2}{n_2} \right) \right\} \tilde{D}^{-1/2} = \tilde{D}^{-1/2} \text{Cov}(\tilde{W}) \tilde{D}^{-1/2} = R.$$

Let

$$Z \sim N(0, R).$$

We next bound the third and fourth Euclidean moments of the summands ξ_ℓ . For $1 \leq i \leq n_1$,

$$\xi_{i,k} = \frac{\sqrt{n_{\text{eff}}}}{n_1} \frac{H_{i,k}^{(1)}}{\sqrt{\tilde{D}_{kk}}}.$$

Since

$$\tilde{D}_{kk} \geq \frac{n_{\text{eff}}}{n_1} \text{Var}(H_{1,k}^{(1)}),$$

we have, for $q = 3, 4$,

$$\mathbb{E}|\xi_{i,k}|^q = \left(\frac{\sqrt{n_{\text{eff}}}}{n_1} \right)^q \frac{\mathbb{E}|H_{i,k}^{(1)}|^q}{\tilde{D}_{kk}^{q/2}} \leq n_1^{-q/2} \frac{\mathbb{E}|H_{i,k}^{(1)}|^q}{\text{Var}(H_{1,k}^{(1)})^{q/2}} \leq M_q n_1^{-q/2}.$$

Similarly, for $1 \leq j \leq n_2$,

$$\mathbb{E}|\xi_{n_1+j,k}|^q \leq M_q n_2^{-q/2}, \quad q = 3, 4.$$

For any $x \in \mathbb{R}^d$,

$$\|x\|_2^3 = \left(\sum_{k=1}^d x_k^2 \right)^{3/2} \leq d^{1/2} \sum_{k=1}^d |x_k|^3,$$

and

$$\|x\|_2^4 = \left(\sum_{k=1}^d x_k^2 \right)^2 \leq d \sum_{k=1}^d |x_k|^4.$$

Therefore,

$$\sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^3 \leq d^{1/2} \sum_{\ell=1}^{n_1+n_2} \sum_{k=1}^d \mathbb{E} |\xi_{\ell,k}|^3.$$

Using the coordinatewise bounds,

$$\sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^3 \leq K_0 d^{3/2} (n_1^{-1/2} + n_2^{-1/2}). \quad (26)$$

Similarly,

$$\sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^4 \leq d \sum_{\ell=1}^{n_1+n_2} \sum_{k=1}^d \mathbb{E} |\xi_{\ell,k}|^4 \leq K_0 d^2 (n_1^{-1} + n_2^{-1}). \quad (27)$$

We now apply Lemma A.1 (the Fang and Koike, 2021 fourth-moment Gaussian approximation on Euclidean balls) to the standardized sum $S_n = \sum_{\ell=1}^{n_1+n_2} \xi_\ell$, which satisfies (24) and $\text{Cov}(S_n) = R$, with reference covariance $\Sigma = R$ (so the centered Gaussian reference $Z \sim N(0, R)$ has the same covariance matrix as S_n). In the notation of Lemma A.1, this corresponds to taking $\Sigma_W = \text{Var}(W_\zeta) = \Sigma = R$. Consequently, the two covariance-comparison terms vanish identically:

$$\delta_0(\Sigma_W, \Sigma) = \sqrt{2 \text{tr}(R) \cdot 2 \|R\|_{\text{op}}} \|R - R\|_{\text{HS}} = 0, \quad \delta'_0(\Sigma_W, \Sigma) = \sum_{j=1}^d |R_{jj} - R_{jj}| = 0.$$

Lemma A.1 therefore reduces, under the present setting, to

$$\sup_{B \in \mathcal{B}} |\mathbb{P}(S_n \in B) - \mathbb{P}(Z \in B)| \leq K \left\{ \kappa(R)^{3/4} \Delta_1^{1/4} + \kappa(R)^{2/3} \Delta_2^{1/3} \right\},$$

where $\mathcal{B}$ denotes the class of Euclidean balls in $\mathbb{R}^d$, and

$$\begin{aligned} \Delta_1 &\leq \|R\|_{\text{HS}} \sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^4 + \|R\|_{\text{op}}^{3/2} \sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^3, \\ \Delta_2 &\leq \|R\|_{\text{op}}^{1/2} \sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^3 + \sum_{\ell=1}^{n_1+n_2} \mathbb{E} \|\xi_\ell\|_2^4. \end{aligned}$$

Substituting (26) and (27) yields

$$\Delta_1 \leq K_0 \left[\|R\|_{\text{HS}} d^2 (n_1^{-1} + n_2^{-1}) + \|R\|_{\text{op}}^{3/2} d^{3/2} (n_1^{-1/2} + n_2^{-1/2}) \right],$$

and

$$\Delta_2 \leq K_0 \left[\|R\|_{\text{op}}^{1/2} d^{3/2} (n_1^{-1/2} + n_2^{-1/2}) + d^2 (n_1^{-1} + n_2^{-1}) \right].$$

Absorbing constants into K , define

$$\delta_1 = \|R\|_{\text{HS}} d^2 (n_1^{-1} + n_2^{-1}) + \|R\|_{\text{op}}^{3/2} d^{3/2} (n_1^{-1/2} + n_2^{-1/2}),$$

and

$$\delta_2 = \|R\|_{\text{op}}^{1/2} d^{3/2} (n_1^{-1/2} + n_2^{-1/2}) + d^2 (n_1^{-1} + n_2^{-1}).$$

Thus

$$\sup_{B \in \mathcal{B}} |\mathbb{P}(S_n \in B) - \mathbb{P}(Z \in B)| \leq K \left\{ \kappa(R)^{3/4} \delta_1^{1/4} + \kappa(R)^{2/3} \delta_2^{1/3} \right\}. \quad (28)$$

Finally, by (25), for any $r \geq 0$,

$$\{\tilde{T}_n \leq r\} = \left\{ \frac{1}{d} \|S_n\|_2^2 \leq r \right\} = \left\{ \|S_n\|_2 \leq \sqrt{dr} \right\},$$

which is a Euclidean ball centered at the origin. Similarly,

$$\{T_Z \leq r\} = \left\{ \|Z\|_2 \leq \sqrt{dr} \right\}.$$

Therefore,

$$\left| \mathbb{P}(\tilde{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq \sup_{B \in \mathcal{B}} |\mathbb{P}(S_n \in B) - \mathbb{P}(Z \in B)|.$$

Taking the supremum over $r \geq 0$ and using (28) gives

$$\sup_{r \geq 0} \left| \mathbb{P}(\tilde{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq K \left\{ \kappa(R)^{3/4} \delta_1^{1/4} + \kappa(R)^{2/3} \delta_2^{1/3} \right\}.$$

This proves the theorem. $\square$

The oracle approximation concerns the ideal statistic $\tilde{T}_n$, which is defined using the true means and the oracle diagonal variance matrix $\tilde{D}$. The statistic used in practice, however, replaces the population means by sample means and replaces $\tilde{D}$ by the feasible diagonal estimator $\hat{D}$. We first show that the diagonal studentizer is uniformly consistent. This result

is then combined with a sample-centering bound to transfer the oracle approximation to the feasible statistic.

Proposition A.1 (Uniform consistency of the diagonal studentizer). Suppose Assumptions 2.1 and 2.2 hold. Let

$$\widehat{D}_{kk} = n_{\text{eff}} \left(\frac{1}{n_1^2} \sum_{i=1}^{n_1} (\widehat{h}_{i,k}^{(1)})^2 + \frac{1}{n_2^2} \sum_{j=1}^{n_2} (\widehat{h}_{j,k}^{(2)})^2 \right), \quad 1 \leq k \leq d.$$

Then there exists a constant $K_0 > 0$, depending only on the constants in Assumptions 2.1 and 2.2, such that

$$\max_{1 \leq k \leq d} \mathbb{E} \left[\left(\frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right)^2 \right] \leq \frac{K_0}{n_0}, \quad n_0 = n_1 \wedge n_2. \quad (29)$$

Consequently, for every $\eta \in (0, 1)$,

$$\mathbb{P} \left(\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right| > K_0 \sqrt{\frac{d}{n_0 \eta}} \right) \leq \eta. \quad (30)$$

Proof of Proposition A.1. We prove the result for a fixed coordinate $k = k(a, b)$, $1 \leq a \leq b \leq p$. The bound is uniform in k , and the final high-probability statement follows from a union bound.

We first consider the first sample. Let

$$G_i = U_{ia} U_{ib}, \quad \mu_G = \mathbb{E} G_i, \quad H_i = G_i - \mu_G, \quad \sigma_{1,k}^2 = \text{Var}(G_i) = \mathbb{E} H_i^2.$$

Let

$$\bar{G} = \frac{1}{n_1} \sum_{i=1}^{n_1} G_i, \quad \bar{H} = \frac{1}{n_1} \sum_{i=1}^{n_1} H_i.$$

Define the oracle sample variance of the covariance-coordinate variable by

$$\check{\sigma}_{1,k}^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} (G_i - \bar{G})^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} (H_i - \bar{H})^2.$$

Since

$$\check{\sigma}_{1,k}^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} H_i^2 - \bar{H}^2,$$

we have

$$\check{\sigma}_{1,k}^2 - \sigma_{1,k}^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} (H_i^2 - \sigma_{1,k}^2) - \bar{H}^2.$$

Therefore,

$$\mathbb{E} \left(\check{\sigma}_{1,k}^2 - \sigma_{1,k}^2 \right)^2 \leq 2\mathbb{E} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} (H_i^2 - \sigma_{1,k}^2) \right]^2 + 2\mathbb{E} \bar{H}^4.$$

By independence,

$$\mathbb{E} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} (H_i^2 - \sigma_{1,k}^2) \right]^2 = \frac{1}{n_1} \text{Var}(H_i^2) \leq \frac{1}{n_1} \mathbb{E} H_i^4.$$

Assumption 2.1 implies

$$\mathbb{E} H_i^4 \leq K_0,$$

uniformly over k . Moreover,

$$\mathbb{E} \bar{H}^4 \leq K_0 n_1^{-2} \mathbb{E} H_i^4 \leq \frac{K_0}{n_1^2}.$$

Hence

$$\mathbb{E} \left(\check{\sigma}_{1,k}^2 - \sigma_{1,k}^2 \right)^2 \leq \frac{K_0}{n_1}.$$

We now compare the feasible residuals with the oracle residuals. Recall that

$$\hat{h}_{i,k}^{(1)} = (U_{ia} - \bar{U}_a)(U_{ib} - \bar{U}_b) - \hat{\Sigma}_{1,ab},$$

where

$$\hat{\Sigma}_{1,ab} = \frac{1}{n_1} \sum_{\ell=1}^{n_1} (U_{\ell a} - \bar{U}_a)(U_{\ell b} - \bar{U}_b).$$

Expanding the centered product gives

$$(U_{ia} - \bar{U}_a)(U_{ib} - \bar{U}_b) = U_{ia}U_{ib} - \bar{U}_aU_{ib} - \bar{U}_bU_{ia} + \bar{U}_a\bar{U}_b.$$

Also,

$$\hat{\Sigma}_{1,ab} = \bar{G} - \bar{U}_a\bar{U}_b.$$

Therefore,

$$\hat{h}_{i,k}^{(1)} = G_i - \bar{G} - \bar{U}_a(U_{ib} - \bar{U}_b) - \bar{U}_b(U_{ia} - \bar{U}_a).$$

Write

$$\hat{h}_{i,k}^{(1)} = (G_i - \bar{G}) + R_i,$$

where

$$R_i = -\bar{U}_a(U_{ib} - \bar{U}_b) - \bar{U}_b(U_{ia} - \bar{U}_a).$$

Define

$$\hat{\sigma}_{1,k}^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} (\hat{h}_{i,k}^{(1)})^2.$$

Then

$$\hat{\sigma}_{1,k}^2 - \check{\sigma}_{1,k}^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} \{2(G_i - \bar{G})R_i + R_i^2\}.$$

By Cauchy's inequality,

$$|\hat{\sigma}_{1,k}^2 - \check{\sigma}_{1,k}^2| \leq 2 \left[\frac{1}{n_1} \sum_{i=1}^{n_1} (G_i - \bar{G})^2 \right]^{1/2} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} R_i^2 \right]^{1/2} + \frac{1}{n_1} \sum_{i=1}^{n_1} R_i^2.$$

Under Assumption 2.1, the fourth moment of $G_i = U_{ia}U_{ib}$ is uniformly bounded. Hence

$$\mathbb{E} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} (G_i - \bar{G})^2 \right]^2 \leq K_0.$$

Next,

$$R_i^2 \leq 2\bar{U}_a^2(U_{ib} - \bar{U}_b)^2 + 2\bar{U}_b^2(U_{ia} - \bar{U}_a)^2.$$

Thus

$$\frac{1}{n_1} \sum_{i=1}^{n_1} R_i^2 \leq 2\bar{U}_a^2 \frac{1}{n_1} \sum_{i=1}^{n_1} (U_{ib} - \bar{U}_b)^2 + 2\bar{U}_b^2 \frac{1}{n_1} \sum_{i=1}^{n_1} (U_{ia} - \bar{U}_a)^2.$$

The coordinate eighth-moment bound implies

$$\mathbb{E} \bar{U}_a^4 \leq \frac{K_0}{n_1^2},$$

and

$$\mathbb{E} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} (U_{ib} - \bar{U}_b)^2 \right]^2 \leq K_0,$$

uniformly over a, b . Hence

$$\mathbb{E} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} R_i^2 \right)^2 \leq \frac{K_0}{n_1^2}.$$

Combining the preceding displays yields

$$\mathbb{E} \left(\hat{\sigma}_{1,k}^2 - \check{\sigma}_{1,k}^2 \right)^2 \leq \frac{K_0}{n_1}.$$

Therefore,

$$\mathbb{E} \left(\hat{\sigma}_{1,k}^2 - \sigma_{1,k}^2 \right)^2 \leq \frac{K_0}{n_1}.$$

The same argument applied to the second sample gives

$$\mathbb{E} \left(\hat{\sigma}_{2,k}^2 - \sigma_{2,k}^2 \right)^2 \leq \frac{K_0}{n_2},$$

where

$$\sigma_{2,k}^2 = \text{Var}(V_{ja} V_{jb}), \quad \hat{\sigma}_{2,k}^2 = \frac{1}{n_2} \sum_{j=1}^{n_2} (\hat{h}_{j,k}^{(2)})^2.$$

Now observe that

$$\widehat{D}_{kk} = n_{\text{eff}} \left(\frac{\hat{\sigma}_{1,k}^2}{n_1} + \frac{\hat{\sigma}_{2,k}^2}{n_2} \right),$$

whereas

$$\widetilde{D}_{kk} = n_{\text{eff}} \left(\frac{\sigma_{1,k}^2}{n_1} + \frac{\sigma_{2,k}^2}{n_2} \right).$$

Therefore,

$$\widehat{D}_{kk} - \widetilde{D}_{kk} = n_{\text{eff}} \left[\frac{\hat{\sigma}_{1,k}^2 - \sigma_{1,k}^2}{n_1} + \frac{\hat{\sigma}_{2,k}^2 - \sigma_{2,k}^2}{n_2} \right].$$

Using $(x + y)^2 \leq 2x^2 + 2y^2$, we get

$$\mathbb{E}(\widehat{D}_{kk} - \widetilde{D}_{kk})^2 \leq 2n_{\text{eff}}^2 \left[\frac{1}{n_1^2} \mathbb{E}(\hat{\sigma}_{1,k}^2 - \sigma_{1,k}^2)^2 + \frac{1}{n_2^2} \mathbb{E}(\hat{\sigma}_{2,k}^2 - \sigma_{2,k}^2)^2 \right].$$

Hence

$$\mathbb{E}(\widehat{D}_{kk} - \widetilde{D}_{kk})^2 \leq K_0 n_{\text{eff}}^2 \left(\frac{1}{n_1^3} + \frac{1}{n_2^3} \right).$$

By Assumption 2.2,

$$\widetilde{D}_{kk} = n_{\text{eff}} \left(\frac{\sigma_{1,k}^2}{n_1} + \frac{\sigma_{2,k}^2}{n_2} \right) \geq n_{\text{eff}} c_H \left(\frac{1}{n_1} + \frac{1}{n_2} \right) = c_H.$$

Thus

$$\mathbb{E} \left[\left(\frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right)^2 \right] \leq K_0 n_{\text{eff}}^2 \left(\frac{1}{n_1^3} + \frac{1}{n_2^3} \right).$$

Since

$$n_{\text{eff}} \leq n_0 = n_1 \wedge n_2,$$

we have

$$n_{\text{eff}}^2 \left(\frac{1}{n_1^3} + \frac{1}{n_2^3} \right) \leq \frac{K_0}{n_0}.$$

Therefore,

$$\max_{1 \leq k \leq d} \mathbb{E} \left[\left(\frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right)^2 \right] \leq \frac{K_0}{n_0}.$$

This proves (29). Finally, for any $t > 0$, Markov's inequality and the union bound give

$$\mathbb{P} \left(\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right| > t \right) \leq \sum_{k=1}^d \frac{\mathbb{E} \left[\left(\widehat{D}_{kk} / \widetilde{D}_{kk} - 1 \right)^2 \right]}{t^2} \leq \frac{Cd}{n_0 t^2}.$$

Taking

$$t = K_0 \sqrt{\frac{d}{n_0 \eta}}$$

with K_0 sufficiently large yields

$$\mathbb{P} \left(\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right| > K_0 \sqrt{\frac{d}{n_0 \eta}} \right) \leq \eta.$$

which proves (30). The proposition follows. $\square$

Next we state the proof of Proposition 2.1.

Proof of Proposition 2.1. Throughout the proof, we work in the transformed model and suppress the superscript (A). Recall that

$$U_i = X_i - \mu_1, \quad V_j = Y_j - \mu_2.$$

Let

$$\widehat{T}_n = \frac{1}{d} \left\| \widehat{D}^{-1/2} \widehat{W} \right\|_2^2, \quad \widetilde{T}_n = \frac{1}{d} \left\| \widetilde{D}^{-1/2} \widetilde{W} \right\|_2^2.$$

We decompose the difference between the feasible statistic and the oracle statistic into the sample-centering error and the diagonal-studentization error.

Define

$$a = d^{-1/2} \widetilde{D}^{-1/2} \widetilde{W}.$$

Then

$$\|a\|_2^2 = \widetilde{T}_n.$$

Also define

$$b = d^{-1/2} \widetilde{D}^{-1/2} (\widehat{W} - \widetilde{W}).$$

Thus

$$d^{-1/2} \widetilde{D}^{-1/2} \widehat{W} = a + b.$$

We first control $\|a\|_2$. Let

$$S_n = \widetilde{D}^{-1/2} \widetilde{W}.$$

Then

$$\widetilde{T}_n = \frac{1}{d} \|S_n\|_2^2.$$

As in the proof of Theorem 2.1, under Assumptions 2.1 and 2.2, the standardized covariance-coordinate summands have uniformly bounded fourth moments. Consequently,

$$\max_{1 \leq k \leq d} \mathbb{E} S_{n,k}^4 \leq K_0.$$

Therefore, by Cauchy–Schwarz,

$$\begin{aligned} \mathbb{E} \widetilde{T}_n^2 &= \mathbb{E} \left(\frac{1}{d} \sum_{k=1}^d S_{n,k}^2 \right)^2 = \frac{1}{d^2} \sum_{k=1}^d \sum_{\ell=1}^d \mathbb{E} (S_{n,k}^2 S_{n,\ell}^2) \\ &\leq \frac{1}{d^2} \sum_{k=1}^d \sum_{\ell=1}^d \left(\mathbb{E} S_{n,k}^4 \right)^{1/2} \left(\mathbb{E} S_{n,\ell}^4 \right)^{1/2} \leq K_0. \end{aligned}$$

Hence, by Markov's inequality,

$$\begin{aligned}\mathbb{P}(\|a\|_2 > B_T(\eta)) &= \mathbb{P}\left(\sqrt{\tilde{T}_n} > K_0\eta^{-1/4}\right) \\ &= \mathbb{P}\left(\tilde{T}_n^2 > K_0^4\eta^{-1}\right) \leq \frac{K_0}{K_0^4\eta^{-1}} \leq \frac{\eta}{3},\end{aligned}$$

after increasing the constant K_0 if necessary. Thus, with probability at least $1 - \eta/3$,

$$\|a\|_2 \leq B_T(\eta),$$

where

$$B_T(\eta) = K_0\eta^{-1/4}.$$

We next control the sample-centering error b . Let

$$\bar{U} = \frac{1}{n_1} \sum_{i=1}^{n_1} U_i, \quad \bar{V} = \frac{1}{n_2} \sum_{j=1}^{n_2} V_j.$$

The oracle covariance matrices are

$$\tilde{\Sigma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} U_i U_i^\top, \quad \tilde{\Sigma}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} V_j V_j^\top.$$

The feasible covariance matrices are

$$\hat{\Sigma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} (U_i - \bar{U})(U_i - \bar{U})^\top,$$

and

$$\hat{\Sigma}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} (V_j - \bar{V})(V_j - \bar{V})^\top.$$

Using the identity

$$\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z})(Z_i - \bar{Z})^\top = \frac{1}{n} \sum_{i=1}^n Z_i Z_i^\top - \bar{Z} \bar{Z}^\top,$$

we obtain

$$\hat{\Sigma}_1 - \tilde{\Sigma}_1 = -\bar{U} \bar{U}^\top, \quad \hat{\Sigma}_2 - \tilde{\Sigma}_2 = -\bar{V} \bar{V}^\top.$$

Therefore,

$$\widehat{W} - \widetilde{W} = \sqrt{n_{\text{eff}}} \text{vech} \{ -\bar{U}\bar{U}^\top + \bar{V}\bar{V}^\top \}.$$

By Assumption 2.2, the oracle diagonal studentizer is uniformly nondegenerate. Indeed, for $k = k(a, b)$,

$$\widetilde{D}_{kk} = n_{\text{eff}} \left\{ \frac{\text{Var}(U_{ia}U_{ib})}{n_1} + \frac{\text{Var}(V_{ja}V_{jb})}{n_2} \right\} \geq n_{\text{eff}} c_H \left(\frac{1}{n_1} + \frac{1}{n_2} \right) = c_H.$$

Hence

$$\|\widetilde{D}^{-1/2}\|_{\text{op}} \leq c_H^{-1/2}.$$

Moreover, for every symmetric matrix A ,

$$\|\text{vech}(A)\|_2 \leq \|A\|_{\text{F}}.$$

Thus

$$\|\widetilde{D}^{-1/2}(\widehat{W} - \widetilde{W})\|_2 \leq K_0 \sqrt{n_{\text{eff}}} \|\bar{U}\bar{U}^\top + \bar{V}\bar{V}^\top\|_{\text{F}}.$$

Using

$$\|uu^\top\|_{\text{F}} = \|u\|_2^2,$$

we get

$$\|\widetilde{D}^{-1/2}(\widehat{W} - \widetilde{W})\|_2 \leq K_0 \sqrt{n_{\text{eff}}} (\|\bar{U}\|_2^2 + \|\bar{V}\|_2^2).$$

Therefore,

$$\|b\|_2 = d^{-1/2} \|\widetilde{D}^{-1/2}(\widehat{W} - \widetilde{W})\|_2 \leq K_0 \frac{\sqrt{n_{\text{eff}}}}{\sqrt{d}} (\|\bar{U}\|_2^2 + \|\bar{V}\|_2^2).$$

We now bound the transformed sample means. By Lemma A.5,

$$\mathbb{E}\|\bar{U}\|_2^4 \leq K_0 \left(\frac{p}{n_1} \right)^2, \quad \mathbb{E}\|\bar{V}\|_2^4 \leq K_0 \left(\frac{p}{n_2} \right)^2.$$

It follows that

$$\mathbb{E}(\|\bar{U}\|_2^2 + \|\bar{V}\|_2^2)^2 \leq 2\mathbb{E}\|\bar{U}\|_2^4 + 2\mathbb{E}\|\bar{V}\|_2^4$$

$$\leq K_0 \left[\left(\frac{p}{n_1} \right)^2 + \left(\frac{p}{n_2} \right)^2 \right] \leq K_0 \left(\frac{p}{n_1} + \frac{p}{n_2} \right)^2.$$

By Markov's inequality, with probability at least $1 - \eta/3$,

$$\|\bar{U}\|_2^2 + \|\bar{V}\|_2^2 \leq K_0 \eta^{-1/2} \left(\frac{p}{n_1} + \frac{p}{n_2} \right).$$

Consequently, with probability at least $1 - \eta/3$,

$$\|b\|_2 \leq K_0 \frac{\sqrt{n_{\text{eff}}}}{\sqrt{d}} \eta^{-1/2} \left(\frac{p}{n_1} + \frac{p}{n_2} \right) = B_W(\eta).$$

We next control the diagonal-studentization error. Define

$$r_D = \max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right|.$$

By the studentizer concentration bound (30), after increasing the constant K_0 if necessary,

$$\mathbb{P} \left(r_D > K_0 \sqrt{\frac{d}{n_0 \eta}} \right) \leq \frac{\eta}{3}.$$

Therefore, with probability at least $1 - \eta/3$,

$$r_D \leq B_D(\eta),$$

where

$$B_D(\eta) = K_0 \left(\frac{d}{n_0 \eta} \right)^{1/2}.$$

Define the event

$$\mathcal{E}_\eta = \{\|a\|_2 \leq B_T(\eta)\} \cap \{\|b\|_2 \leq B_W(\eta)\} \cap \{r_D \leq B_D(\eta)\}. \quad (31)$$

By the preceding bounds and the union bound,

$$\mathbb{P}(\mathcal{E}_\eta) \geq 1 - \eta. \quad (32)$$

We now prove the deterministic perturbation bound on the event $\mathcal{E}_\eta$ in (31). Define

$$E = \widetilde{D}^{1/2} \widehat{D}^{-1} \widetilde{D}^{1/2}.$$

Since E is diagonal,

$$E_{kk} = \frac{\widetilde{D}_{kk}}{\widehat{D}_{kk}}.$$

Moreover,

$$\widehat{T}_n = \frac{1}{d} \widehat{W}^\top \widehat{D}^{-1} \widehat{W} = (a+b)^\top E(a+b),$$

whereas

$$\widetilde{T}_n = a^\top a.$$

On $\mathcal{E}_\eta$, if

$$B_D(\eta) \leq \frac{1}{2},$$

then

$$\frac{1}{2} \leq \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} \leq \frac{3}{2}$$

for every $1 \leq k \leq d$. Hence $\widehat{D}$ is positive definite, and

$$|E_{kk} - 1| = \left| \frac{\widetilde{D}_{kk}}{\widehat{D}_{kk}} - 1 \right| = \frac{|\widehat{D}_{kk}/\widetilde{D}_{kk} - 1|}{\widehat{D}_{kk}/\widetilde{D}_{kk}} \leq 2r_D.$$

Thus

$$\|E - I_d\|_{\text{op}} \leq 2r_D.$$

We decompose

$$\begin{aligned} \widehat{T}_n - \widetilde{T}_n &= (a+b)^\top E(a+b) - a^\top a \\ &= (a+b)^\top (E - I_d)(a+b) + (\|a+b\|_2^2 - \|a\|_2^2). \end{aligned}$$

Therefore,

$$|\widehat{T}_n - \widetilde{T}_n| \leq |(a+b)^\top (E - I_d)(a+b)| + \left| \|a+b\|_2^2 - \|a\|_2^2 \right|.$$

The first term satisfies

$$|(a+b)^\top (E - I_d)(a+b)| \leq \|E - I_d\|_{\text{op}} \|a+b\|_2^2 \leq 2r_D (\|a\|_2 + \|b\|_2)^2.$$

For the second term,

$$\|a + b\|_2^2 - \|a\|_2^2 = 2a^\top b + \|b\|_2^2,$$

so

$$\left| \|a + b\|_2^2 - \|a\|_2^2 \right| \leq 2\|a\|_2\|b\|_2 + \|b\|_2^2.$$

Combining the two inequalities gives

$$|\widehat{T}_n - \widetilde{T}_n| \leq 2r_D(\|a\|_2 + \|b\|_2)^2 + 2\|a\|_2\|b\|_2 + \|b\|_2^2. \quad (33)$$

On $\mathcal{E}_\eta$,

$$\|a\|_2 \leq B_T(\eta), \quad \|b\|_2 \leq B_W(\eta), \quad r_D \leq B_D(\eta).$$

Hence (33) gives

$$|\widehat{T}_n - \widetilde{T}_n| \leq 2B_D(\eta)\{B_T(\eta) + B_W(\eta)\}^2 + 2B_T(\eta)B_W(\eta) + B_W^2(\eta).$$

By the definition of $\Delta(\eta)$,

$$|\widehat{T}_n - \widetilde{T}_n| \leq \Delta(\eta)$$

on $\mathcal{E}_\eta$. Since (32) holds, the first claim follows.

It remains to verify the explicit rate in the final statement. Take

$$\eta_n = n_0^{-1/8}.$$

Then

$$B_T(\eta_n) = K_0\eta_n^{-1/4} = K_0n_0^{1/32},$$

and

$$B_D(\eta_n) = K_0 \left(\frac{d}{n_0\eta_n} \right)^{1/2} = K_0 \left(\frac{d}{n_0^{7/8}} \right)^{1/2}.$$

Moreover, since

$$n_{\text{eff}} \leq n_0, \quad n_1, n_2 \geq n_0,$$

we have

$$\frac{p}{n_1} + \frac{p}{n_2} \leq \frac{2p}{n_0}.$$

Because

$$d = \frac{p(p+1)}{2},$$

we also have

$$\frac{p}{\sqrt{d}} \leq K_0.$$

Therefore,

$$B_W(\eta_n) = K_0 \frac{\sqrt{n_{\text{eff}}}}{\sqrt{d}} n_0^{1/16} \left(\frac{p}{n_1} + \frac{p}{n_2} \right) \leq K_0 \frac{\sqrt{n_0}}{\sqrt{d}} n_0^{1/16} \frac{p}{n_0} \leq K_0 n_0^{-7/16}.$$

Substituting these bounds into

$$\Delta(\eta_n) = 2B_D(\eta_n)\{B_T(\eta_n) + B_W(\eta_n)\}^2 + 2B_T(\eta_n)B_W(\eta_n) + B_W^2(\eta_n),$$

we obtain

$$\begin{aligned} \Delta(\eta_n) &\leq K_0 B_D(\eta_n) B_T^2(\eta_n) + K_0 B_D(\eta_n) B_T(\eta_n) B_W(\eta_n) \\ &\quad + K_0 B_D(\eta_n) B_W^2(\eta_n) + K_0 B_T(\eta_n) B_W(\eta_n) + K_0 B_W^2(\eta_n). \end{aligned}$$

The first leading term is

$$B_D(\eta_n) B_T^2(\eta_n) = K_0 \left(\frac{d}{n_0^{7/8}} \right)^{1/2} n_0^{1/16} = K_0 \left(\frac{d}{n_0^{3/4}} \right)^{1/2}.$$

The second leading term is

$$B_T(\eta_n) B_W(\eta_n) \leq K_0 n_0^{1/32} n_0^{-7/16} = K_0 n_0^{-13/32}.$$

For the remaining terms, using

$$B_D(\eta_n) \leq \frac{1}{2}$$

and $B_T(\eta_n) \geq 1$ for large n_0 , we have

$$B_D(\eta_n) B_T(\eta_n) B_W(\eta_n) \leq K_0 B_T(\eta_n) B_W(\eta_n) \leq K_0 n_0^{-13/32},$$

and

$$B_D(\eta_n)B_W^2(\eta_n) \leq K_0 B_W^2(\eta_n) \leq K_0 n_0^{-7/8}.$$

Also,

$$B_W^2(\eta_n) \leq K_0 n_0^{-7/8}.$$

Since

$$n_0^{-7/8} \leq n_0^{-13/32}$$

for $n_0 \geq 1$, all remaining terms are dominated by

$$K_0 \left[\left(\frac{d}{n_0^{3/4}} \right)^{1/2} + n_0^{-13/32} \right].$$

Therefore,

$$\Delta(\eta_n) \leq K_0 \left[\left(\frac{d}{n_0^{3/4}} \right)^{1/2} + n_0^{-13/32} \right].$$

Since the first part of the proposition gives

$$\mathbb{P}(|\widehat{T}_n - \widetilde{T}_n| \leq \Delta(\eta_n)) \geq 1 - \eta_n = 1 - n_0^{-1/8},$$

we conclude that, with probability at least $1 - n_0^{-1/8}$,

$$|\widehat{T}_n - \widetilde{T}_n| \leq K_0 \left[\left(\frac{d}{n_0^{3/4}} \right)^{1/2} + n_0^{-13/32} \right].$$

This completes the proof. □

Proof of Corollary 2.2. Let

$$F_{\widehat{T}}(r) = \mathbb{P}(\widehat{T}_n \leq r), \quad F_{\widetilde{T}}(r) = \mathbb{P}(\widetilde{T}_n \leq r),$$

and

$$G(r) = \mathbb{P}(T_Z \leq r).$$

By Theorem 2.1,

$$\sup_{r \geq 0} |F_{\widetilde{T}}(r) - G(r)| \leq \varepsilon_{\text{or}},$$

where

$$\varepsilon_{\text{or}} = K \left\{ \kappa(R)^{3/4} \delta_1^{1/4} + \kappa(R)^{2/3} \delta_2^{1/3} \right\}.$$

Next, by Proposition 2.1, if

$$B_D(\eta) \leq \frac{1}{2},$$

then the event

$$\mathcal{E}_\eta = \left\{ |\widehat{T}_n - \widetilde{T}_n| \leq \Delta(\eta) \right\}$$

satisfies

$$\mathbb{P}(\mathcal{E}_\eta) \geq 1 - \eta.$$

We first derive an upper bound for

$$F_{\widehat{T}}(r) - G(r).$$

For any fixed $r \geq 0$,

$$\mathbb{P}(\widehat{T}_n \leq r) = \mathbb{P}(\widehat{T}_n \leq r, \mathcal{E}_\eta) + \mathbb{P}(\widehat{T}_n \leq r, \mathcal{E}_\eta^c).$$

Since

$$\mathbb{P}(\mathcal{E}_\eta^c) \leq \eta,$$

we have

$$\mathbb{P}(\widehat{T}_n \leq r) \leq \mathbb{P}(\widehat{T}_n \leq r, \mathcal{E}_\eta) + \eta.$$

On the event $\mathcal{E}_\eta$,

$$|\widehat{T}_n - \widetilde{T}_n| \leq \Delta(\eta).$$

Therefore,

$$\widehat{T}_n \leq r \implies \widetilde{T}_n \leq r + \Delta(\eta).$$

Hence

$$\mathbb{P}(\widehat{T}_n \leq r) \leq \mathbb{P}(\widetilde{T}_n \leq r + \Delta(\eta)) + \eta.$$

Using the oracle Gaussian approximation,

$$\mathbb{P}(\tilde{T}_n \leq r + \Delta(\eta)) \leq \mathbb{P}(T_Z \leq r + \Delta(\eta)) + \varepsilon_{\text{or}}.$$

Thus

$$\mathbb{P}(\hat{T}_n \leq r) \leq \mathbb{P}(T_Z \leq r + \Delta(\eta)) + \varepsilon_{\text{or}} + \eta.$$

By the definition of the anti-concentration modulus

$$\omega_Z(a) = \sup_{s \geq 0} \mathbb{P}(s < T_Z \leq s + a),$$

we have

$$\mathbb{P}(T_Z \leq r + \Delta(\eta)) \leq \mathbb{P}(T_Z \leq r) + \omega_Z\{\Delta(\eta)\}.$$

Therefore,

$$\mathbb{P}(\hat{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \leq \varepsilon_{\text{or}} + \eta + \omega_Z\{\Delta(\eta)\}.$$

We next derive the reverse inequality. On $\mathcal{E}_\eta$,

$$\tilde{T}_n \leq r - \Delta(\eta) \implies \hat{T}_n \leq r.$$

If $r - \Delta(\eta) < 0$, the event

$$\{\tilde{T}_n \leq r - \Delta(\eta)\}$$

is interpreted as the empty event. Hence, for every $r \geq 0$,

$$\mathbb{P}(\hat{T}_n \leq r) \geq \mathbb{P}(\tilde{T}_n \leq r - \Delta(\eta), \mathcal{E}_\eta).$$

Therefore,

$$\mathbb{P}(\hat{T}_n \leq r) \geq \mathbb{P}(\tilde{T}_n \leq r - \Delta(\eta)) - \mathbb{P}(\mathcal{E}_\eta^c).$$

Since

$$\mathbb{P}(\mathcal{E}_\eta^c) \leq \eta,$$

we get

$$\mathbb{P}(\hat{T}_n \leq r) \geq \mathbb{P}(\tilde{T}_n \leq r - \Delta(\eta)) - \eta.$$

Using the oracle Gaussian approximation again,

$$\mathbb{P}(\tilde{T}_n \leq r - \Delta(\eta)) \geq \mathbb{P}(T_Z \leq r - \Delta(\eta)) - \varepsilon_{\text{or}}.$$

Thus

$$\mathbb{P}(\hat{T}_n \leq r) \geq \mathbb{P}(T_Z \leq r - \Delta(\eta)) - \varepsilon_{\text{or}} - \eta.$$

It follows that

$$\mathbb{P}(T_Z \leq r) - \mathbb{P}(\hat{T}_n \leq r) \leq \mathbb{P}(T_Z \leq r) - \mathbb{P}(T_Z \leq r - \Delta(\eta)) + \varepsilon_{\text{or}} + \eta.$$

By the definition of ω_Z ,

$$\mathbb{P}(T_Z \leq r) - \mathbb{P}(T_Z \leq r - \Delta(\eta)) \leq \omega_Z\{\Delta(\eta)\}.$$

Therefore,

$$\mathbb{P}(T_Z \leq r) - \mathbb{P}(\hat{T}_n \leq r) \leq \varepsilon_{\text{or}} + \eta + \omega_Z\{\Delta(\eta)\}.$$

Combining the two one-sided bounds, we obtain, for every $r \geq 0$,

$$\left| \mathbb{P}(\hat{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq \varepsilon_{\text{or}} + \eta + \omega_Z\{\Delta(\eta)\}.$$

Taking the supremum over $r \geq 0$ gives

$$\sup_{r \geq 0} \left| \mathbb{P}(\hat{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq \varepsilon_{\text{or}} + \eta + \omega_Z\{\Delta(\eta)\}.$$

It remains to use the Gaussian quadratic-form anti-concentration bound. Under Assumption 2.3,

$$T_Z = \frac{1}{d} \|Z\|_2^2, \quad Z \sim N(0, R).$$

Applying Lemma A.2 (the Götze–Naumov–Spokoiny–Ulyanov anti-concentration bound, as recorded in Fang and Koike, 2021, Lemma 5.10) with $\Sigma = R$ and $\mu = 0$, we obtain, for any $\varepsilon > 0$,

$$\sup_{a \geq 0} \mathbb{P}(a \leq \|Z\|_2^2 \leq a + \varepsilon) \leq K\kappa(R)\varepsilon.$$

Translating to the rescaled statistic $T_Z = d^{-1}\|Z\|_2^2$ via the change of variable $a = rd$, $\varepsilon = ad$, gives

$$\omega_Z(a) = \sup_{r \geq 0} \mathbb{P}(r < T_Z \leq r + a) = \sup_{a' \geq 0} \mathbb{P}(a' < \|Z\|_2^2 \leq a' + ad) \leq K\kappa(R) \cdot ad = K\kappa(R)da.$$

Taking $a = \Delta(\eta)$, we obtain

$$\omega_Z\{\Delta(\eta)\} \leq K\kappa(R)d\Delta(\eta).$$

Hence

$$\sup_{r \geq 0} \left| \mathbb{P}(\widehat{T}_n \leq r) - \mathbb{P}(T_Z \leq r) \right| \leq \varepsilon_{\text{or}} + \eta + K\kappa(R)d\Delta(\eta).$$

This completes the proof. $\square$

The multiplier bootstrap statistic is conditionally Gaussian given the data, with conditional covariance matrix $\widehat{R}$. To compare the bootstrap law with the oracle Gaussian reference $T_Z = d^{-1}\|Z\|_2^2$, $Z \sim N(0, R)$, we need to control the discrepancy between $\widehat{R}$ and R . The next proposition provides a conservative nuclear-norm bound for this discrepancy. The proof is technical and is given in the appendix.

Proposition A.2 (Bootstrap covariance consistency). Suppose Assumptions 2.1 and 2.2 hold. Let

$$\Gamma = \text{Cov}(\widetilde{W}), \quad R = \widetilde{D}^{-1/2}\Gamma\widetilde{D}^{-1/2}.$$

Let

$$\widehat{\Gamma} = n_{\text{eff}} \left\{ \frac{1}{n_1^2} \sum_{i=1}^{n_1} \widehat{h}_i^{(1)} (\widehat{h}_i^{(1)})^\top + \frac{1}{n_2^2} \sum_{j=1}^{n_2} \widehat{h}_j^{(2)} (\widehat{h}_j^{(2)})^\top \right\},$$

and

$$\widehat{R} = \widehat{D}^{-1/2} \widehat{\Gamma} \widehat{D}^{-1/2}.$$

Then there exists a constant $K_R > 0$, depending only on the constants in the assumptions, such that for every $\eta \in (0, 1)$,

$$\mathbb{P} \left(\|\widehat{R} - R\|_{S_1} > K_R \sqrt{\frac{d^3}{n_0 \eta}} \right) \leq \eta, \quad n_0 = n_1 \wedge n_2.$$

Equivalently,

$$\|\widehat{R} - R\|_{S_1} = O_{\mathbb{P}} \left(\sqrt{\frac{d^3}{n_0}} \right).$$

Proof. Throughout the proof, we work in the transformed model and suppress the superscript (A). Let

$$U_i = X_i - \mu_1, \quad V_j = Y_j - \mu_2.$$

Define the oracle covariance-coordinate vectors

$$H_i^{(1)} = \text{vech}\{U_i U_i^\top - \mathbb{E}(U_i U_i^\top)\},$$

and

$$H_j^{(2)} = \text{vech}\{V_j V_j^\top - \mathbb{E}(V_j V_j^\top)\}.$$

Let

$$\Gamma_1 = \text{Cov}(H_i^{(1)}), \quad \Gamma_2 = \text{Cov}(H_j^{(2)}).$$

Then

$$\Gamma = \text{Cov}(\widetilde{W}) = n_{\text{eff}} \left(\frac{\Gamma_1}{n_1} + \frac{\Gamma_2}{n_2} \right).$$

We first control the covariance estimator before diagonal standardization. For the first sample, define

$$\bar{H}^{(1)} = \frac{1}{n_1} \sum_{i=1}^{n_1} H_i^{(1)}$$

and

$$\check{\Gamma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} (H_i^{(1)} - \bar{H}^{(1)})(H_i^{(1)} - \bar{H}^{(1)})^\top.$$

Also define

$$\hat{\Gamma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} \hat{h}_i^{(1)} (\hat{h}_i^{(1)})^\top.$$

The second-sample analogues $\check{\Gamma}_2$ and $\hat{\Gamma}_2$ are defined similarly.

We claim that

$$\mathbb{E} \|\hat{\Gamma}_1 - \Gamma_1\|_{\text{F}}^2 \leq \frac{Cd^2}{n_1}, \quad \mathbb{E} \|\hat{\Gamma}_2 - \Gamma_2\|_{\text{F}}^2 \leq \frac{Cd^2}{n_2}.$$

We prove the first bound; the second is identical.

Write

$$\Gamma_{1,0} = \frac{1}{n_1} \sum_{i=1}^{n_1} H_i^{(1)} (H_i^{(1)})^\top.$$

Then

$$\check{\Gamma}_1 = \Gamma_{1,0} - \bar{H}^{(1)} (\bar{H}^{(1)})^\top.$$

Therefore,

$$\|\check{\Gamma}_1 - \Gamma_1\|_F^2 \leq 2\|\Gamma_{1,0} - \Gamma_1\|_F^2 + 2\|\bar{H}^{(1)} (\bar{H}^{(1)})^\top\|_F^2.$$

Since the $H_i^{(1)}$'s are i.i.d. centered,

$$\mathbb{E}\|\Gamma_{1,0} - \Gamma_1\|_F^2 = \frac{1}{n_1} \mathbb{E}\|H_i^{(1)} (H_i^{(1)})^\top - \Gamma_1\|_F^2.$$

Moreover,

$$\|H_i^{(1)} (H_i^{(1)})^\top\|_F^2 = \|H_i^{(1)}\|_2^4.$$

By Assumption 2.1, for every coordinate k ,

$$\mathbb{E}|H_{i,k}^{(1)}|^4 \leq K_0.$$

Hence

$$\mathbb{E}\|H_i^{(1)}\|_2^4 = \mathbb{E}\left(\sum_{k=1}^d (H_{i,k}^{(1)})^2\right)^2 \leq d \sum_{k=1}^d \mathbb{E}|H_{i,k}^{(1)}|^4 \leq Cd^2.$$

It follows that

$$\mathbb{E}\|\Gamma_{1,0} - \Gamma_1\|_F^2 \leq \frac{Cd^2}{n_1}.$$

Also,

$$\|\bar{H}^{(1)} (\bar{H}^{(1)})^\top\|_F^2 = \|\bar{H}^{(1)}\|_2^4.$$

Using the same fourth-moment calculation for the sample mean of the d -dimensional vector

$H_i^{(1)}$, together with $\mathbb{E}\|H_i^{(1)}\|_2^4 \leq Cd^2$, gives

$$\mathbb{E}\|\bar{H}^{(1)}\|_2^4 \leq \frac{Cd^2}{n_1^2}.$$

Thus

$$\mathbb{E}\|\check{\Gamma}_1 - \Gamma_1\|_F^2 \leq \frac{Cd^2}{n_1}.$$

We next compare the feasible residuals with the oracle centered residuals. Let

$$g_i^{(1)} = H_i^{(1)} - \bar{H}^{(1)}.$$

Then

$$\check{\Gamma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} g_i^{(1)} (g_i^{(1)})^\top.$$

By the sample-centering expansion,

$$\hat{h}_i^{(1)} = g_i^{(1)} + r_i^{(1)},$$

where

$$r_i^{(1)} = \text{vech} \left\{ -\bar{U}(U_i - \bar{U})^\top - (U_i - \bar{U})\bar{U}^\top \right\}.$$

Indeed, this follows from expanding

$$(U_i - \bar{U})(U_i - \bar{U})^\top - \hat{\Sigma}_1$$

and comparing it with

$$U_i U_i^\top - \frac{1}{n_1} \sum_{\ell=1}^{n_1} U_\ell U_\ell^\top.$$

Therefore,

$$\hat{\Gamma}_1 - \check{\Gamma}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} \left\{ g_i^{(1)} (r_i^{(1)})^\top + r_i^{(1)} (g_i^{(1)})^\top + r_i^{(1)} (r_i^{(1)})^\top \right\}.$$

Using

$$\|xy^\top\|_F = \|x\|_2 \|y\|_2,$$

we obtain

$$\|\hat{\Gamma}_1 - \check{\Gamma}_1\|_F \leq 2 \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|g_i^{(1)}\|_2^2 \right)^{1/2} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|r_i^{(1)}\|_2^2 \right)^{1/2} + \frac{1}{n_1} \sum_{i=1}^{n_1} \|r_i^{(1)}\|_2^2.$$

Set

$$L_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} \|g_i^{(1)}\|_2^2, \quad M_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} \|r_i^{(1)}\|_2^2.$$

Then

$$\|\hat{\Gamma}_1 - \check{\Gamma}_1\|_F^2 \leq K_0(L_1 M_1 + M_1^2).$$

We bound L_1 and M_1 . Since

$$\mathbb{E}\|H_i^{(1)}\|_2^4 \leq C d^2,$$

we have

$$\mathbb{E} L_1^2 \leq C d^2.$$

For M_1 , note that

$$\|r_i^{(1)}\|_2 \leq \|-\bar{U}(U_i - \bar{U})^\top - (U_i - \bar{U})\bar{U}^\top\|_F \leq 2\|\bar{U}\|_2\|U_i - \bar{U}\|_2.$$

Therefore,

$$M_1 \leq 4\|\bar{U}\|_2^2 \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|U_i - \bar{U}\|_2^2 \right).$$

By Lemma A.5,

$$\mathbb{E}\|\bar{U}\|_2^8 \leq K_0 \left(\frac{p}{n_1} \right)^4,$$

and

$$\mathbb{E} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|U_i - \bar{U}\|_2^2 \right)^4 \leq C p^4.$$

Consequently, by Cauchy–Schwarz,

$$\mathbb{E} M_1^2 \leq K_0 (\mathbb{E}\|\bar{U}\|_2^8)^{1/2} \left[\mathbb{E} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} \|U_i - \bar{U}\|_2^2 \right)^4 \right]^{1/2} \leq K_0 \frac{p^4}{n_1^2}.$$

Since

$$d = \frac{p(p+1)}{2} \asymp p^2,$$

we have $p^4 \leq C d^2$, and hence

$$\mathbb{E} M_1^2 \leq \frac{C d^2}{n_1^2}.$$

Therefore,

$$\mathbb{E}(L_1 M_1) \leq (\mathbb{E} L_1^2)^{1/2} (\mathbb{E} M_1^2)^{1/2} \leq Cd \cdot \frac{d}{n_1} = \frac{Cd^2}{n_1}.$$

Together with $\mathbb{E} M_1^2 \leq Cd^2/n_1^2$, this gives

$$\mathbb{E} \|\hat{\Gamma}_1 - \check{\Gamma}_1\|_F^2 \leq \frac{Cd^2}{n_1}.$$

Combining this with the bound for $\check{\Gamma}_1 - \Gamma_1$, we conclude that

$$\mathbb{E} \|\hat{\Gamma}_1 - \Gamma_1\|_F^2 \leq \frac{Cd^2}{n_1}.$$

Similarly,

$$\mathbb{E} \|\hat{\Gamma}_2 - \Gamma_2\|_F^2 \leq \frac{Cd^2}{n_2}.$$

Now recall that

$$\hat{\Gamma} = n_{\text{eff}} \left(\frac{\hat{\Gamma}_1}{n_1} + \frac{\hat{\Gamma}_2}{n_2} \right),$$

and

$$\Gamma = n_{\text{eff}} \left(\frac{\Gamma_1}{n_1} + \frac{\Gamma_2}{n_2} \right).$$

Thus

$$\hat{\Gamma} - \Gamma = n_{\text{eff}} \left[\frac{\hat{\Gamma}_1 - \Gamma_1}{n_1} + \frac{\hat{\Gamma}_2 - \Gamma_2}{n_2} \right].$$

Using $(x + y)^2 \leq 2x^2 + 2y^2$,

$$\|\hat{\Gamma} - \Gamma\|_F^2 \leq 2n_{\text{eff}}^2 \left[\frac{\|\hat{\Gamma}_1 - \Gamma_1\|_F^2}{n_1^2} + \frac{\|\hat{\Gamma}_2 - \Gamma_2\|_F^2}{n_2^2} \right].$$

Taking expectations gives

$$\mathbb{E} \|\hat{\Gamma} - \Gamma\|_F^2 \leq K_0 n_{\text{eff}}^2 d^2 \left(\frac{1}{n_1^3} + \frac{1}{n_2^3} \right).$$

Since $n_{\text{eff}} \leq n_0 = n_1 \wedge n_2$,

$$n_{\text{eff}}^2 \left(\frac{1}{n_1^3} + \frac{1}{n_2^3} \right) \leq \frac{K_0}{n_0}.$$

Hence

$$\mathbb{E}\|\hat{\Gamma} - \Gamma\|_{\text{F}}^2 \leq \frac{Cd^2}{n_0}.$$

By Markov's inequality,

$$\mathbb{P}\left(\|\hat{\Gamma} - \Gamma\|_{\text{F}} > K_0 \frac{d}{\sqrt{n_0\eta}}\right) \leq \frac{\eta}{2} \quad (34)$$

after increasing K_0 if necessary.

We now pass from $\hat{\Gamma} - \Gamma$ to $\hat{R} - R$. Let

$$r_D = \max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{kk}}{\widetilde{D}_{kk}} - 1 \right|.$$

By the studentizer concentration bound (30),

$$\mathbb{P}\left(r_D > K_0 \sqrt{\frac{d}{n_0\eta}}\right) \leq \frac{\eta}{2}. \quad (35)$$

On the event $r_D \leq 1/2$, the matrix $\widehat{D}$ is positive definite and

$$\|\widehat{D}^{-1/2}\|_{\text{op}} \leq K_0, \quad \|\widetilde{D}^{-1/2}\|_{\text{op}} \leq K_0.$$

Moreover,

$$\left\| \widehat{D}^{-1/2} \widetilde{D}^{1/2} - I_d \right\|_{\text{op}} \leq Cr_D.$$

Define

$$B = \widehat{D}^{-1/2} \widetilde{D}^{1/2}.$$

Then

$$\widehat{D}^{-1/2} \Gamma \widehat{D}^{-1/2} = B \left(\widetilde{D}^{-1/2} \Gamma \widetilde{D}^{-1/2} \right) B = BRB.$$

Therefore,

$$\widehat{R} - R = \widehat{D}^{-1/2} (\hat{\Gamma} - \Gamma) \widehat{D}^{-1/2} + (BRB - R). \quad (36)$$

For the first term in (36),

$$\left\| \widehat{D}^{-1/2} (\hat{\Gamma} - \Gamma) \widehat{D}^{-1/2} \right\|_{S_1} \leq \|\widehat{D}^{-1/2}\|_{\text{op}}^2 \|\hat{\Gamma} - \Gamma\|_{S_1}.$$

Since

$$\|M\|_{S_1} \leq \sqrt{d} \|M\|_F,$$

we have

$$\left\| \widehat{D}^{-1/2} (\widehat{\Gamma} - \Gamma) \widehat{D}^{-1/2} \right\|_{S_1} \leq K_0 \sqrt{d} \|\widehat{\Gamma} - \Gamma\|_F.$$

For the second term in (36),

$$BRB - R = (B - I_d)RB + R(B - I_d).$$

Hence

$$\|BRB - R\|_{S_1} \leq \|B - I_d\|_{\text{op}} \|R\|_{S_1} \|B\|_{\text{op}} + \|R\|_{S_1} \|B - I_d\|_{\text{op}}.$$

Because $\Gamma = \text{Cov}(\widetilde{W})$ is positive semidefinite and $\widetilde{D}^{-1/2}$ is symmetric, the matrix $R = \widetilde{D}^{-1/2} \Gamma \widetilde{D}^{-1/2}$ is positive semidefinite. Moreover, by definition of $\widetilde{D}$, the diagonal entries of R are all equal to one. Therefore,

$$\|R\|_{S_1} = \text{tr}(R) = d.$$

Also,

$$\|B - I_d\|_{\text{op}} \leq K_0 r_D, \quad \|B\|_{\text{op}} \leq K_0.$$

Therefore,

$$\|BRB - R\|_{S_1} \leq C d r_D.$$

Combining the two bounds,

$$\|\widehat{R} - R\|_{S_1} \leq K_0 \sqrt{d} \|\widehat{\Gamma} - \Gamma\|_F + C d r_D. \tag{37}$$

On the event where

$$\|\widehat{\Gamma} - \Gamma\|_F \leq K_0 \frac{d}{\sqrt{n_0 \eta}}$$

and

$$r_D \leq K_0 \sqrt{\frac{d}{n_0 \eta}},$$

we therefore have, by (37),

$$\|\widehat{R} - R\|_{S_1} \leq K_0 \sqrt{d} \cdot \frac{d}{\sqrt{n_0 \eta}} + Cd \sqrt{\frac{d}{n_0 \eta}} \leq K_0 \sqrt{\frac{d^3}{n_0 \eta}}.$$

The probability of this event is at least $1 - \eta$. Hence

$$\mathbb{P} \left(\|\widehat{R} - R\|_{S_1} > K_R \sqrt{\frac{d^3}{n_0 \eta}} \right) \leq \eta$$

for a sufficiently large constant $K_R > 0$. This proves the proposition. $\square$

Proof of Theorem 2.3. We first identify the conditional distribution of the multiplier statistic. Recall that

$$\widehat{W}^{\text{mb}} = \sqrt{n_{\text{eff}}} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} e_i \hat{h}_i^{(1)} - \frac{1}{n_2} \sum_{j=1}^{n_2} e'_j \hat{h}_j^{(2)} \right\},$$

where, conditionally on the data, the multiplier variables

$$e_1, \dots, e_{n_1}, e'_1, \dots, e'_{n_2}$$

are independent standard normal random variables. Therefore, conditionally on the data,

$$\widehat{W}^{\text{mb}} \mid \mathcal{X} \sim N(0, \widehat{\Gamma}),$$

where

$$\widehat{\Gamma} = n_{\text{eff}} \left\{ \frac{1}{n_1^2} \sum_{i=1}^{n_1} \hat{h}_i^{(1)} (\hat{h}_i^{(1)})^\top + \frac{1}{n_2^2} \sum_{j=1}^{n_2} \hat{h}_j^{(2)} (\hat{h}_j^{(2)})^\top \right\}.$$

Consequently,

$$\widehat{Z} := \widehat{D}^{-1/2} \widehat{W}^{\text{mb}} \mid \mathcal{X} \sim N(0, \widehat{R}),$$

where

$$\widehat{R} = \widehat{D}^{-1/2} \widehat{\Gamma} \widehat{D}^{-1/2}.$$

Thus, conditionally on the data,

$$\widehat{T}_n^{\text{mb}} = \frac{1}{d} \left\| \widehat{D}^{-1/2} \widehat{W}^{\text{mb}} \right\|_2^2 = \frac{1}{d} \|\widehat{Z}\|_2^2.$$

The oracle Gaussian reference is

$$T_Z = \frac{1}{d} \|Z\|_2^2, \quad Z \sim N(0, R),$$

where

$$R = \widetilde{D}^{-1/2} \text{Cov}(\widetilde{W}) \widetilde{D}^{-1/2}.$$

Hence the bootstrap approximation problem reduces to comparing two Gaussian quadratic forms:

$$\frac{1}{d} \|\widehat{Z}\|_2^2, \quad \frac{1}{d} \|Z\|_2^2,$$

with covariance matrices $\widehat{R}$ and R , respectively.

By Proposition A.2, there exists a constant $K_R > 0$ such that, for every $\eta_2 \in (0, 1)$,

$$\mathbb{P} \left(\|\widehat{R} - R\|_{S_1} > K_R \sqrt{\frac{d^3}{n_0 \eta_2}} \right) \leq \eta_2.$$

Define the event

$$\mathcal{B}_{\eta_2} = \left\{ \|\widehat{R} - R\|_{S_1} \leq K_R \sqrt{\frac{d^3}{n_0 \eta_2}} \right\}.$$

Then

$$\mathbb{P}(\mathcal{B}_{\eta_2}) \geq 1 - \eta_2.$$

We now show that on $\mathcal{B}_{\eta_2}$, the Gaussian reference geometry of $\widehat{R}$ is controlled by that of R . Since

$$\|\widehat{R} - R\|_{\text{op}} \leq \|\widehat{R} - R\|_{S_1}$$

and

$$\|\widehat{R} - R\|_{\text{F}} \leq \|\widehat{R} - R\|_{S_1},$$

on $\mathcal{B}_{\eta_2}$ we have

$$\|\widehat{R} - R\|_{\text{F}} \leq K_R \sqrt{\frac{d^3}{n_0 \eta_2}}.$$

Let

$$\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_d \geq 0$$

be the eigenvalues of $\widehat{R}$, and define

$$\widehat{\Lambda}_s = \left(\sum_{\ell=s}^d \hat{\lambda}_\ell^2 \right)^{1/2}, \quad s = 1, 2.$$

By Lemma A.4 (the Hoffman–Wielandt inequality applied to $\widehat{R}$ and R),

$$\left[\sum_{\ell=1}^d (\hat{\lambda}_\ell - \lambda_\ell)^2 \right]^{1/2} \leq \|\widehat{R} - R\|_F.$$

Therefore, on $\mathcal{B}_{\eta_2}$,

$$|\widehat{\Lambda}_s - \Lambda_s| \leq \|\widehat{R} - R\|_F \leq K_R \sqrt{\frac{d^3}{n_0 \eta_2}}, \quad s = 1, 2.$$

By the assumed geometry stability condition,

$$K_R \sqrt{\frac{d^3}{n_0 \eta_2}} \leq \frac{1}{2} \Lambda_2(R).$$

Since $\Lambda_1(R) \geq \Lambda_2(R)$, the same bound also implies

$$K_R \sqrt{\frac{d^3}{n_0 \eta_2}} \leq \frac{1}{2} \Lambda_1(R).$$

Hence, on $\mathcal{B}_{\eta_2}$,

$$\widehat{\Lambda}_1 \geq \Lambda_1(R) - K_R \sqrt{\frac{d^3}{n_0 \eta_2}} \geq \frac{1}{2} \Lambda_1(R),$$

and

$$\widehat{\Lambda}_2 \geq \Lambda_2(R) - K_R \sqrt{\frac{d^3}{n_0 \eta_2}} \geq \frac{1}{2} \Lambda_2(R).$$

In particular,

$$\widehat{\Lambda}_2 > 0$$

on $\mathcal{B}_{\eta_2}$. Moreover,

$$\kappa(\widehat{R}) = (\widehat{\Lambda}_1 \widehat{\Lambda}_2)^{-1/2} \leq 2(\Lambda_1(R) \Lambda_2(R))^{-1/2} = 2\kappa(R).$$

We now apply Lemma A.3 (the Gaussian comparison inequality for squared norms). Conditional on the data and on the event $\mathcal{B}_{\eta_2}$, both covariance matrices $\widehat{R}$ and R satisfy the required nondegeneracy condition ($\Lambda_2(\cdot) > 0$ by the preceding step). Lemma A.3 therefore gives, after the rescaling $\widehat{T}_n^{\text{mb}} = d^{-1}\|\widehat{Z}\|_2^2$ and $T_Z = d^{-1}\|Z\|_2^2$,

$$\sup_{r \geq 0} \left| \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \leq r \mid \mathcal{X} \right) - \mathbb{P}(T_Z \leq r) \right| \leq K \{ \kappa(\widehat{R}) + \kappa(R) \} \|\widehat{R} - R\|_{S_1}.$$

Using

$$\kappa(\widehat{R}) \leq 2\kappa(R)$$

and

$$\|\widehat{R} - R\|_{S_1} \leq K_R \sqrt{\frac{d^3}{n_0 \eta_2}}$$

on $\mathcal{B}_{\eta_2}$, we obtain

$$\sup_{r \geq 0} \left| \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \leq r \mid \mathcal{X} \right) - \mathbb{P}(T_Z \leq r) \right| \leq K \kappa(R) \sqrt{\frac{d^3}{n_0 \eta_2}}.$$

By the definition

$$\rho_{\text{mb}}(\eta_2) = K \kappa(R) \sqrt{\frac{d^3}{n_0 \eta_2}},$$

this becomes

$$\sup_{r \geq 0} \left| \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \leq r \mid \mathcal{X} \right) - \mathbb{P}(T_Z \leq r) \right| \leq \rho_{\text{mb}}(\eta_2)$$

on $\mathcal{B}_{\eta_2}$.

Since

$$\mathbb{P}(\mathcal{B}_{\eta_2}) \geq 1 - \eta_2,$$

the asserted high-probability bootstrap approximation follows. $\square$

Proof of Corollary 2.4. Let

$$F_n(r) = \mathbb{P}_{H_0}(\widehat{T}_n \leq r)$$

be the unconditional distribution function of the feasible statistic under H_0 , and let

$$G(r) = \mathbb{P}(T_Z \leq r)$$

be the distribution function of the oracle Gaussian reference

$$T_Z = \frac{1}{d} \|Z\|_2^2, \quad Z \sim N(0, R).$$

Also define the conditional bootstrap distribution function

$$\widehat{F}_{\text{mb}}(r) = \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \leq r \mid \mathcal{X} \right).$$

By Corollary 2.2 and the assumptions of Corollary 2.4, there exists a deterministic error $\varepsilon_n \rightarrow 0$, given by the bound in (11), such that

$$\sup_{r \geq 0} |F_n(r) - G(r)| \leq \varepsilon_n. \quad (38)$$

Next, by Theorem 2.3 and the assumptions of Corollary 2.4, there exists a deterministic error $\rho_n \rightarrow 0$, given by the bound in (16), such that with probability tending to one,

$$\sup_{r \geq 0} |\widehat{F}_{\text{mb}}(r) - G(r)| \leq \rho_n. \quad (39)$$

Let

$$\mathcal{B}_n = \left\{ \sup_{r \geq 0} |\widehat{F}_{\text{mb}}(r) - G(r)| \leq \rho_n \right\}. \quad (40)$$

Then

$$\mathbb{P}(\mathcal{B}_n) \rightarrow 1.$$

Let $\tau_n = \mathbb{P}(\mathcal{B}_n^c)$. Then $\tau_n \rightarrow 0$.

We first verify that the oracle Gaussian reference has well-behaved quantiles. Since

$$T_Z = \frac{1}{d} \|Z\|_2^2, \quad Z \sim N(0, R),$$

we may diagonalize R and write

$$T_Z \stackrel{d}{=} \frac{1}{d} \sum_{\ell=1}^d \lambda_{\ell} \xi_{\ell}^2,$$

where $\xi_1, \dots, \xi_d$ are independent standard normal random variables and $\lambda_1, \dots, \lambda_d$ are the eigenvalues of R . By Assumption 2.3,

$$\Lambda_2 = \left(\sum_{\ell=2}^d \lambda_{\ell}^2 \right)^{1/2} > 0.$$

Thus the Gaussian quadratic form is nondegenerate. In particular, G is continuous on $[0, \infty)$. Moreover, for every $u \in (0, 1)$, if

$$q_u = \inf\{r \geq 0 : G(r) \geq u\},$$

then

$$G(q_u) = u.$$

Let

$$\hat{q}_u = \inf\{r \geq 0 : \widehat{F}_{\text{mb}}(r) \geq u\}$$

be the conditional bootstrap u -quantile. By definition,

$$\hat{c}_{1-\alpha}^{\text{mb}} = \hat{q}_{1-\alpha}.$$

On the event $\mathcal{B}_n$ in (40), the Kolmogorov bound (39) implies the quantile sandwich

$$q_{1-\alpha-\rho_n} \leq \hat{c}_{1-\alpha}^{\text{mb}} \leq q_{1-\alpha+\rho_n}, \tag{41}$$

for all sufficiently large n , since $\rho_n \rightarrow 0$.

Indeed, if

$$r < q_{1-\alpha-\rho_n},$$

then

$$G(r) < 1 - \alpha - \rho_n.$$

Hence, on $\mathcal{B}_n$,

$$\widehat{F}_{\text{mb}}(r) \leq G(r) + \rho_n < 1 - \alpha.$$

Therefore r cannot reach the bootstrap $(1 - \alpha)$ -quantile, giving

$$q_{1-\alpha-\rho_n} \leq \widehat{c}_{1-\alpha}^{\text{mb}}.$$

Similarly, since

$$G(q_{1-\alpha+\rho_n}) = 1 - \alpha + \rho_n,$$

we have, on $\mathcal{B}_n$,

$$\widehat{F}_{\text{mb}}(q_{1-\alpha+\rho_n}) \geq G(q_{1-\alpha+\rho_n}) - \rho_n = 1 - \alpha.$$

Therefore

$$\widehat{c}_{1-\alpha}^{\text{mb}} \leq q_{1-\alpha+\rho_n}.$$

We now bound the rejection probability from above. By the left inequality in (41), on $\mathcal{B}_n$,

$$\{\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}\} \subset \{\widehat{T}_n > q_{1-\alpha-\rho_n}\}$$

Hence

$$\mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \leq \mathbb{P}_{H_0}(\widehat{T}_n > q_{1-\alpha-\rho_n}) + \mathbb{P}(\mathcal{B}_n^c).$$

Since $\mathbb{P}(\mathcal{B}_n^c) = \tau_n$,

$$\mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \leq 1 - F_n(q_{1-\alpha-\rho_n}) + \tau_n.$$

Using (38), we obtain

$$1 - F_n(q_{1-\alpha-\rho_n}) \leq 1 - G(q_{1-\alpha-\rho_n}) + \varepsilon_n.$$

Since

$$G(q_{1-\alpha-\rho_n}) = 1 - \alpha - \rho_n,$$

we get

$$1 - G(q_{1-\alpha-\rho_n}) = \alpha + \rho_n.$$

Therefore,

$$\mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \leq \alpha + \rho_n + \varepsilon_n + \tau_n. \quad (42)$$

Because

$$\rho_n \rightarrow 0, \quad \varepsilon_n \rightarrow 0, \quad \tau_n \rightarrow 0,$$

we have

$$\limsup_{n \rightarrow \infty} \mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \leq \alpha.$$

We next derive the lower bound. By the right inequality in (41), on $\mathcal{B}_n$,

$$\{\widehat{T}_n > q_{1-\alpha+\rho_n}\} \subset \{\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}\}$$

Therefore,

$$\mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \geq \mathbb{P}_{H_0}(\widehat{T}_n > q_{1-\alpha+\rho_n}) - \mathbb{P}(\mathcal{B}_n^c).$$

Since $\mathbb{P}(\mathcal{B}_n^c) = \tau_n$,

$$\mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \geq 1 - F_n(q_{1-\alpha+\rho_n}) - \tau_n.$$

Again using (38), we get

$$1 - F_n(q_{1-\alpha+\rho_n}) \geq 1 - G(q_{1-\alpha+\rho_n}) - \varepsilon_n.$$

Since

$$G(q_{1-\alpha+\rho_n}) = 1 - \alpha + \rho_n,$$

we have

$$1 - G(q_{1-\alpha+\rho_n}) = \alpha - \rho_n.$$

Thus

$$\mathbb{P}_{H_0}(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}}) \geq \alpha - \rho_n - \varepsilon_n - \tau_n. \quad (43)$$

Letting $n \rightarrow \infty$, and using

$$\rho_n \rightarrow 0, \quad \varepsilon_n \rightarrow 0, \quad \tau_n \rightarrow 0,$$

gives

$$\liminf_{n \rightarrow \infty} \mathbb{P}_{H_0} \left(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}} \right) \geq \alpha.$$

Combining the upper and lower bounds, we conclude that

$$\mathbb{P}_{H_0} \left(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}} \right) \rightarrow \alpha.$$

This proves the corollary. □

Proof of Theorem 2.5. Throughout the proof, we work in the transformed model and suppress the superscript (A) . Under H_1 , write

$$\Delta_\Sigma = \text{vech}\{A(\Sigma_1 - \Sigma_2)A^\top\}, \quad \theta_n = \widetilde{D}^{-1/2}\Delta_\Sigma.$$

The oracle covariance-difference vector can be decomposed as

$$\widetilde{W} = \sqrt{n_{\text{eff}}}\Delta_\Sigma + \widetilde{W}_0,$$

where

$$\widetilde{W}_0 = \sqrt{n_{\text{eff}}} \left\{ \text{vech}(\widetilde{\Sigma}_1 - \Sigma_1) - \text{vech}(\widetilde{\Sigma}_2 - \Sigma_2) \right\}$$

is centered. Define

$$S_n = \widetilde{D}^{-1/2}\widetilde{W}_0.$$

Then

$$\widetilde{D}^{-1/2}\widetilde{W} = \sqrt{n_{\text{eff}}}\theta_n + S_n.$$

By construction, the diagonal entries of $\text{Cov}(S_n)$ are equal to one. Hence

$$\mathbb{E}_{H_1} \left(\frac{1}{d} \|S_n\|_2^2 \right) = \frac{1}{d} \text{tr}\{\text{Cov}(S_n)\} = 1.$$

Therefore,

$$\frac{1}{\sqrt{d}} \|S_n\|_2 = O_{\mathbb{P}}(1).$$

By the primitive alternative condition, there exists $\mathcal{J}_n \subset \{1, \dots, d\}$, with $|\mathcal{J}_n| = m_n$, such that

$$\min_{k \in \mathcal{J}_n} |\theta_{n,k}| \geq a_n.$$

Consequently,

$$\frac{1}{d} \|\sqrt{n_{\text{eff}}} \theta_n\|_2^2 = \frac{n_{\text{eff}}}{d} \|\theta_n\|_2^2 \geq n_{\text{eff}} \frac{m_n}{d} a_n^2 = g_n.$$

Since $g_n \rightarrow \infty$, we have

$$\frac{1}{\sqrt{d}} \|\sqrt{n_{\text{eff}}} \theta_n\|_2 \geq \sqrt{g_n} \rightarrow \infty.$$

Combining this deterministic lower bound with $d^{-1/2} \|S_n\|_2 = O_{\mathbb{P}}(1)$, the triangle inequality gives

$$\frac{1}{\sqrt{d}} \|\widetilde{D}^{-1/2} \widetilde{W}\|_2 = \frac{1}{\sqrt{d}} \|\sqrt{n_{\text{eff}}} \theta_n + S_n\|_2 \geq \sqrt{g_n} - O_{\mathbb{P}}(1).$$

Thus

$$\widetilde{T}_n = \frac{1}{d} \|\widetilde{D}^{-1/2} \widetilde{W}\|_2^2 \rightarrow \infty$$

in probability under H_1 .

It remains to transfer this divergence from the oracle statistic to the feasible statistic. Let

$$E_n = \widetilde{D}^{-1/2} (\widehat{W} - \widetilde{W}).$$

The same sample-centering argument used in the proof of Proposition 2.1 gives, under Assumption 2.1,

$$\frac{1}{\sqrt{d}} \|E_n\|_2 = O_{\mathbb{P}} \left[\sqrt{\frac{n_{\text{eff}}}{d}} \left(\frac{p}{n_1} + \frac{p}{n_2} \right) \right].$$

By the assumed growth condition,

$$\sqrt{\frac{n_{\text{eff}}}{d}} \left(\frac{p}{n_1} + \frac{p}{n_2} \right) = o(\sqrt{g_n}),$$

we obtain

$$\frac{1}{\sqrt{d}} \|E_n\|_2 = o_{\mathbb{P}}(\sqrt{g_n}).$$

Next, by the coordinatewise studentizer consistency established in the sample-to-oracle analysis,

$$\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_k}{\widetilde{D}_k} - 1 \right| = O_{\mathbb{P}} \left(\sqrt{\frac{d}{n_0}} \right) = o_{\mathbb{P}}(1),$$

where the last equality follows from $d/n_0 \rightarrow 0$. Hence, with probability tending to one,

$$\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_k}{\widetilde{D}_k} - 1 \right| \leq \frac{1}{2}.$$

On this event,

$$\frac{\widetilde{D}_k}{\widehat{D}_k} \geq \frac{2}{3} \quad \text{for all } k = 1, \dots, d.$$

Therefore,

$$\widehat{T}_n = \frac{1}{d} \widehat{W}^\top \widehat{D}^{-1} \widehat{W} = \frac{1}{d} \sum_{k=1}^d \frac{\widetilde{D}_k}{\widehat{D}_k} [\widetilde{D}^{-1/2} \widehat{W}]_k^2 \geq \frac{2}{3} \frac{1}{d} \|\widetilde{D}^{-1/2} \widehat{W}\|_2^2.$$

Since

$$\widetilde{D}^{-1/2} \widehat{W} = \widetilde{D}^{-1/2} \widetilde{W} + E_n = \sqrt{n_{\text{eff}}} \theta_n + S_n + E_n,$$

another application of the triangle inequality yields

$$\frac{1}{\sqrt{d}} \|\widetilde{D}^{-1/2} \widehat{W}\|_2 \geq \sqrt{g_n} - \frac{1}{\sqrt{d}} \|S_n\|_2 - \frac{1}{\sqrt{d}} \|E_n\|_2.$$

The second term is $O_{\mathbb{P}}(1)$, while the third term is $o_{\mathbb{P}}(\sqrt{g_n})$. Since $g_n \rightarrow \infty$,

$$\frac{1}{\sqrt{d}} \|\widetilde{D}^{-1/2} \widehat{W}\|_2 \rightarrow \infty$$

in probability. Consequently,

$$\widehat{T}_n \rightarrow \infty$$

in probability under H_1 .

It remains only to show that the bootstrap critical value does not diverge. Conditionally on the data, the multiplier-bootstrap statistic has the form

$$\widehat{T}_n^{\text{mb}} = \frac{1}{d} \|\widehat{D}^{-1/2} \widehat{W}^{\text{mb}}\|_2^2.$$

Let

$$\widehat{Z} = \widehat{D}^{-1/2} \widehat{W}^{\text{mb}}.$$

Then, conditionally on the data,

$$\widehat{Z} \mid \mathcal{X} \sim N(0, \widehat{R}),$$

where $\widehat{R}$ has diagonal entries equal to one by construction of $\widehat{D}$. Hence

$$\text{tr}(\widehat{R}) = d$$

and

$$\mathbb{E}_{\xi} \left(\widehat{T}_n^{\text{mb}} \mid \mathcal{X} \right) = \frac{1}{d} \text{tr}(\widehat{R}) = 1.$$

By conditional Markov's inequality, for any $M > 0$,

$$\mathbb{P}_{\xi} \left(\widehat{T}_n^{\text{mb}} > M \mid \mathcal{X} \right) \leq \frac{1}{M}.$$

Choose $M > \alpha^{-1}$. Then

$$\mathbb{P}_{\xi} \left(\widehat{T}_n^{\text{mb}} \leq M \mid \mathcal{X} \right) \geq 1 - \frac{1}{M} > 1 - \alpha.$$

Therefore, by the definition of the conditional $(1 - \alpha)$ -quantile,

$$\widehat{c}_{1-\alpha}^{\text{mb}} \leq M$$

with probability tending to one. Thus

$$\widehat{c}_{1-\alpha}^{\text{mb}} = O_{\mathbb{P}}(1).$$

We have shown that

$$\widehat{T}_n \rightarrow \infty \quad \text{in probability,} \quad \widehat{c}_{1-\alpha}^{\text{mb}} = O_{\mathbb{P}}(1).$$

It follows that

$$\mathbb{P}_{H_1} \left(\widehat{T}_n > \widehat{c}_{1-\alpha}^{\text{mb}} \right) \rightarrow 1.$$

Finally, the right-tail multiplier-bootstrap p -value is

$$p_n^{\text{mb}} = \mathbb{P}_\xi \left(\widehat{T}_n^{\text{mb}} \geq \widehat{T}_n \mid \mathcal{X} \right).$$

By conditional Markov's inequality,

$$p_n^{\text{mb}} \leq \frac{\mathbb{E}_\xi(\widehat{T}_n^{\text{mb}} \mid \mathcal{X})}{\widehat{T}_n} = \frac{1}{\widehat{T}_n}.$$

Since $\widehat{T}_n \rightarrow \infty$ in probability, we conclude that

$$p_n^{\text{mb}} \rightarrow 0$$

in probability. This completes the proof. $\square$

Proof of Proposition 2.2. We work under H_0 and suppress the superscript (A) . Recall that

$$\widetilde{W} = \sqrt{n_{\text{eff}}}(\bar{H}_1 - \bar{H}_2), \quad W_{2|1} = -\sqrt{n_2}\bar{H}_2.$$

Since the two samples are independent,

$$\text{Cov}(\widetilde{W}) = \frac{n_{\text{eff}}}{n_1}\Gamma_1 + \frac{n_{\text{eff}}}{n_2}\Gamma_2.$$

Therefore,

$$\widetilde{D} = \alpha_n D_1 + \beta_n D_2,$$

where

$$\alpha_n = \frac{n_{\text{eff}}}{n_1} = \frac{n_2}{n_1 + n_2}, \quad \beta_n = \frac{n_{\text{eff}}}{n_2} = \frac{n_1}{n_1 + n_2}.$$

Writing

$$\varepsilon_n = \frac{n_2}{n_1},$$

we have

$$\alpha_n = \frac{\varepsilon_n}{1 + \varepsilon_n} = O(\varepsilon_n), \quad \beta_n = \frac{1}{1 + \varepsilon_n} = 1 + O(\varepsilon_n).$$

Hence

$$\frac{n_{\text{eff}}}{n_2} = \beta_n = 1 + O(\varepsilon_n).$$

By Assumption 2.2,

$$D_{g,k} \geq c_H \quad \text{for all } g = 1, 2, \quad k = 1, \dots, d.$$

By Assumption 2.1, there exists a constant $C < \infty$ such that

$$D_{g,k} \leq C \quad \text{for all } g = 1, 2, \quad k = 1, \dots, d.$$

Indeed, for a covariance coordinate $k = (a, b)$,

$$D_{g,k} = \text{Var}(Q_{ga}Q_{gb}) \leq \mathbb{E}(Q_{ga}^2 Q_{gb}^2) \leq (\mathbb{E}|Q_{ga}|^4)^{1/2} (\mathbb{E}|Q_{gb}|^4)^{1/2} \leq C,$$

where Q_g denotes a generic transformed centered observation from group g , that is, $Q_1 = U_i$ and $Q_2 = V_j$.

Consequently,

$$\max_{1 \leq k \leq d} \left| \frac{\tilde{D}_k}{D_{2,k}} - 1 \right| = \max_{1 \leq k \leq d} \left| \alpha_n \left(\frac{D_{1,k}}{D_{2,k}} - 1 \right) \right| \leq C \alpha_n = O(\varepsilon_n).$$

It follows that

$$\|\tilde{D}^{-1/2} D_2^{1/2}\|_{\text{op}} = O(1), \quad \|\tilde{D}^{-1/2} D_2^{1/2} - I_d\|_{\text{op}} = O(\varepsilon_n).$$

Next,

$$\tilde{W} - W_{2|1} = \sqrt{n_{\text{eff}}} \bar{H}_1 + (\sqrt{n_2} - \sqrt{n_{\text{eff}}}) \bar{H}_2.$$

Since $\bar{H}_1$ and $\bar{H}_2$ are independent and centered,

$$\begin{aligned} \mathbb{E} \|D_2^{-1/2} (\tilde{W} - W_{2|1})\|_2^2 &= \frac{n_{\text{eff}}}{n_1} \text{tr} (D_2^{-1/2} \Gamma_1 D_2^{-1/2}) \\ &\quad + \frac{(\sqrt{n_2} - \sqrt{n_{\text{eff}}})^2}{n_2} \text{tr} (D_2^{-1/2} \Gamma_2 D_2^{-1/2}). \end{aligned}$$

Using the uniform bounds on $D_{2,k}$ and $\Gamma_{1,kk}$,

$$\text{tr} (D_2^{-1/2} \Gamma_1 D_2^{-1/2}) = \sum_{k=1}^d \frac{\Gamma_{1,kk}}{D_{2,k}} \leq Cd.$$

Moreover,

$$\mathrm{tr} \left(D_2^{-1/2} \Gamma_2 D_2^{-1/2} \right) = \sum_{k=1}^d \frac{\Gamma_{2,kk}}{D_{2,k}} = d.$$

Therefore,

$$\frac{1}{d} \mathbb{E} \left\| D_2^{-1/2} (\widetilde{W} - W_{2|1}) \right\|_2^2 \leq C \alpha_n + (1 - \sqrt{\beta_n})^2.$$

Since

$$\alpha_n = O(\varepsilon_n), \quad 1 - \sqrt{\beta_n} = O(\varepsilon_n),$$

we obtain

$$\frac{1}{d} \mathbb{E} \left\| D_2^{-1/2} (\widetilde{W} - W_{2|1}) \right\|_2^2 \leq C \varepsilon_n.$$

By Markov's inequality,

$$\frac{1}{\sqrt{d}} \left\| D_2^{-1/2} (\widetilde{W} - W_{2|1}) \right\|_2 = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

Now decompose

$$\widetilde{D}^{-1/2} \widetilde{W} - D_2^{-1/2} W_{2|1} = \widetilde{D}^{-1/2} (\widetilde{W} - W_{2|1}) + (\widetilde{D}^{-1/2} - D_2^{-1/2}) W_{2|1}.$$

For the first term,

$$\begin{aligned} \frac{1}{\sqrt{d}} \left\| \widetilde{D}^{-1/2} (\widetilde{W} - W_{2|1}) \right\|_2 &\leq \left\| \widetilde{D}^{-1/2} D_2^{1/2} \right\|_{\mathrm{op}} \frac{1}{\sqrt{d}} \left\| D_2^{-1/2} (\widetilde{W} - W_{2|1}) \right\|_2 \\ &= O_{\mathbb{P}}(\varepsilon_n^{1/2}). \end{aligned}$$

For the second term,

$$(\widetilde{D}^{-1/2} - D_2^{-1/2}) W_{2|1} = (\widetilde{D}^{-1/2} D_2^{1/2} - I_d) D_2^{-1/2} W_{2|1}.$$

Furthermore,

$$\mathbb{E} \left[\frac{1}{d} \left\| D_2^{-1/2} W_{2|1} \right\|_2^2 \right] = \frac{1}{d} \mathrm{tr} \left(D_2^{-1/2} \Gamma_2 D_2^{-1/2} \right) = 1.$$

Thus

$$\frac{1}{\sqrt{d}} \left\| D_2^{-1/2} W_{2|1} \right\|_2 = O_{\mathbb{P}}(1),$$

and the second term is $O_{\mathbb{P}}(\varepsilon_n)$ after normalization by $d^{-1/2}$. Combining the two terms gives

$$\frac{1}{\sqrt{d}} \left\| \widetilde{D}^{-1/2} \widetilde{W} - D_2^{-1/2} W_{2|1} \right\|_2 = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

Finally, let

$$A_n = \widetilde{D}^{-1/2} \widetilde{W}, \quad B_n = D_2^{-1/2} W_{2|1}.$$

We have shown that

$$d^{-1/2} \|A_n - B_n\|_2 = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

Also,

$$d^{-1/2} \|B_n\|_2 = O_{\mathbb{P}}(1),$$

and hence

$$d^{-1/2} \|A_n\|_2 = O_{\mathbb{P}}(1).$$

Using

$$\left| \|A_n\|_2^2 - \|B_n\|_2^2 \right| \leq \|A_n - B_n\|_2 (\|A_n\|_2 + \|B_n\|_2),$$

we obtain

$$|\widetilde{T}_n - T_{2|1}^{\text{or}}| = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

This proves the proposition. □

Proof of Corollary 2.6. We prove that the feasible simplified statistic is close to its oracle version. Write

$$\widehat{W}_{2|1} = \sqrt{n_2} \text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_2), \quad W_{2|1} = -\sqrt{n_2} \bar{H}_2.$$

Under H_0 , $\Sigma_1 = \Sigma_2$, and

$$W_{2|1} = \sqrt{n_2} \text{vech}(\Sigma_1 - \widetilde{\Sigma}_2).$$

Therefore,

$$\widehat{W}_{2|1} - W_{2|1} = \sqrt{n_2} \text{vech} \left[(\widehat{\Sigma}_1 - \Sigma_1) + (\widetilde{\Sigma}_2 - \widehat{\Sigma}_2) \right].$$

Using the sample-centering identity

$$\widehat{\Sigma}_g - \widetilde{\Sigma}_g = -(\bar{X}_g - \mu_g)(\bar{X}_g - \mu_g)^\top, \quad g = 1, 2,$$

we obtain

$$\widehat{W}_{2|1} - W_{2|1} = \sqrt{n_2} \text{vech} \left[(\widetilde{\Sigma}_1 - \Sigma_1) - (\bar{X} - \mu_1)(\bar{X} - \mu_1)^\top + (\bar{Y} - \mu_2)(\bar{Y} - \mu_2)^\top \right].$$

We first control this vector after $D_2^{-1/2}$ -standardization. The leading term is

$$\sqrt{n_2} \text{vech}(\widetilde{\Sigma}_1 - \Sigma_1) = \sqrt{n_2} \bar{H}_1.$$

As in the proof of Proposition 2.2,

$$\mathbb{E} \left[\frac{1}{d} \left\| D_2^{-1/2} \sqrt{n_2} \bar{H}_1 \right\|_2^2 \right] = \frac{n_2}{n_1} \frac{1}{d} \text{tr}(D_2^{-1/2} \Gamma_1 D_2^{-1/2}) \leq C \varepsilon_n.$$

Hence

$$\frac{1}{\sqrt{d}} \left\| D_2^{-1/2} \sqrt{n_2} \bar{H}_1 \right\|_2 = O_{\mathbb{P}}(\varepsilon_n^{1/2}).$$

For the sample-centering terms, use

$$\left\| \text{vech}(aa^\top) \right\|_2 \leq \|aa^\top\|_F = \|a\|_2^2.$$

By Lemma A.5,

$$\|\bar{X} - \mu_1\|_2^2 = O_{\mathbb{P}}\left(\frac{p}{n_1}\right), \quad \|\bar{Y} - \mu_2\|_2^2 = O_{\mathbb{P}}\left(\frac{p}{n_2}\right).$$

Since $D_{2,k} \geq c_H$ and $d = p(p+1)/2$, we have

$$\frac{\sqrt{n_2}}{\sqrt{d}} \left\| D_2^{-1/2} \text{vech}\{(\bar{X} - \mu_1)(\bar{X} - \mu_1)^\top\} \right\|_2 = O_{\mathbb{P}}\left(\frac{\sqrt{n_2}}{\sqrt{d}} \frac{p}{n_1}\right) = O_{\mathbb{P}}\left(\frac{\sqrt{n_2}}{n_1}\right),$$

and

$$\frac{\sqrt{n_2}}{\sqrt{d}} \left\| D_2^{-1/2} \text{vech}\{(\bar{Y} - \mu_2)(\bar{Y} - \mu_2)^\top\} \right\|_2 = O_{\mathbb{P}}\left(\frac{\sqrt{n_2}}{\sqrt{d}} \frac{p}{n_2}\right) = O_{\mathbb{P}}(n_2^{-1/2}).$$

The first term is $O_{\mathbb{P}}(\varepsilon_n^{1/2})$, and the second term is absorbed by $O_{\mathbb{P}}\{(d/n_2)^{1/2}\}$. Therefore,

$$\frac{1}{\sqrt{d}} \|D_2^{-1/2}(\widehat{W}_{2|1} - W_{2|1})\|_2 = O_{\mathbb{P}} \left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}} \right).$$

Next, the one-sample version of the diagonal studentizer consistency gives

$$\max_{1 \leq k \leq d} \left| \frac{\widehat{D}_{2,kk}}{D_{2,k}} - 1 \right| = O_{\mathbb{P}} \left(\sqrt{\frac{d}{n_2}} \right).$$

Since $d/n_2 \rightarrow 0$, with probability tending to one,

$$\|\widehat{D}_2^{-1/2} D_2^{1/2} - I_d\|_{\text{op}} = O_{\mathbb{P}} \left(\sqrt{\frac{d}{n_2}} \right), \quad \|\widehat{D}_2^{-1/2} D_2^{1/2}\|_{\text{op}} = O_{\mathbb{P}}(1).$$

Now decompose

$$\widehat{D}_2^{-1/2} \widehat{W}_{2|1} - D_2^{-1/2} W_{2|1} = \widehat{D}_2^{-1/2} D_2^{1/2} D_2^{-1/2} (\widehat{W}_{2|1} - W_{2|1}) + (\widehat{D}_2^{-1/2} D_2^{1/2} - I_d) D_2^{-1/2} W_{2|1}.$$

The first term is

$$O_{\mathbb{P}} \left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}} \right)$$

after normalization by $d^{-1/2}$. For the second term, recall from the proof of Proposition 2.2 that

$$\frac{1}{\sqrt{d}} \|D_2^{-1/2} W_{2|1}\|_2 = O_{\mathbb{P}}(1).$$

Hence the second term is

$$O_{\mathbb{P}} \left(\sqrt{\frac{d}{n_2}} \right)$$

after normalization by $d^{-1/2}$. Therefore,

$$\frac{1}{\sqrt{d}} \|\widehat{D}_2^{-1/2} \widehat{W}_{2|1} - D_2^{-1/2} W_{2|1}\|_2 = O_{\mathbb{P}} \left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}} \right).$$

Let

$$A_n = \widehat{D}_2^{-1/2} \widehat{W}_{2|1}, \quad B_n = D_2^{-1/2} W_{2|1}.$$

The preceding display gives

$$d^{-1/2}\|A_n - B_n\|_2 = O_{\mathbb{P}}\left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}}\right).$$

Moreover,

$$d^{-1/2}\|B_n\|_2 = O_{\mathbb{P}}(1),$$

and the same is true for A_n . Therefore,

$$|\widehat{T}_{2|1} - T_{2|1}^{\text{or}}| \leq \frac{1}{\sqrt{d}}\|A_n - B_n\|_2 \left(\frac{1}{\sqrt{d}}\|A_n\|_2 + \frac{1}{\sqrt{d}}\|B_n\|_2 \right),$$

which yields

$$\widehat{T}_{2|1} - T_{2|1}^{\text{or}} = O_{\mathbb{P}}\left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}}\right).$$

Finally, by the triangle inequality,

$$|\widehat{T}_n - \widehat{T}_{2|1}| \leq |\widehat{T}_n - \widetilde{T}_n| + |\widetilde{T}_n - T_{2|1}^{\text{or}}| + |T_{2|1}^{\text{or}} - \widehat{T}_{2|1}|.$$

Using Proposition 2.2 and the first part of this corollary gives

$$|\widehat{T}_n - \widehat{T}_{2|1}| \leq |\widehat{T}_n - \widetilde{T}_n| + O_{\mathbb{P}}\left(\varepsilon_n^{1/2} + \sqrt{\frac{d}{n_2}}\right).$$

If $\widehat{T}_n - \widetilde{T}_n = o_{\mathbb{P}}(1)$, $\varepsilon_n \rightarrow 0$, and $d/n_2 \rightarrow 0$, then

$$\widehat{T}_n - \widehat{T}_{2|1} = o_{\mathbb{P}}(1).$$

This completes the proof. □

B Additional Simulation Diagnostic

For completeness, we report the simulation diagnostic for the simplified implementation in highly unbalanced designs. This diagnostic is not used in the main simulation section, which focuses on empirical size across dimensions and power under dense alternatives. The

Table 5: Unbalanced-sample diagnostic with $p = 50$ and $R = 1000$. The full and simplified columns report empirical size. The relative statistic difference is $|T_{\text{full}} - T_{\text{simp}}|/\{(|T_{\text{full}}| + |T_{\text{simp}}|)/2\}$, averaged over Monte Carlo replications; agreement is the percentage of replications in which the two procedures make the same rejection decision.

covariance	n_1	n_2	n_1/n_2	full	simplified	mean rel. Δ_T	agreement
Toeplitz(0.8)	5000	100	50	0.049	0.056	2.4%	99.1%

setting uses $p = 50$, $n_1 = 5000$, $n_2 = 100$, and a Toeplitz null covariance with $\rho = 0.8$. The purpose is to compare the full two-sample statistic with the simplified statistic that treats the larger sample as a stable reference covariance.

The simplified statistic is close to the full statistic in this highly unbalanced setting, and both rejection rates are near the nominal level. This supports the asymptotic simplification as a useful diagnostic approximation when the information in the larger sample is comparatively stable.

We also examined finite-sample size as the sample-size ratio moves toward the balanced case. The design fixes $n_2 = 60$, uses $(n_1, n_2) \in \{(300, 60), (180, 60), (120, 60), (90, 60), (60, 60)\}$, and sets $\rho_0 = 0.5$. For each $p \in \{100, 200\}$, the simulation averages over equicorrelation and Toeplitz covariance families and over Gaussian, standardized centered χ_1^2 , and standardized Laplace innovations. The full setting-level output, including all benchmark methods, is retained in the formal simulation archive. Table 6 reports the average empirical sizes for the proposed target-transformed tests and the identity implementation.

The ratio-grid diagnostic is consistent with the balanced and mildly unbalanced size results when $n_1/n_2 \leq 2$. For stronger imbalance, especially $n_1/n_2 \in \{3, 5\}$ with $n_2 = 60$, the target-transformed procedures become liberal on average. We therefore treat this experiment as a finite-sample warning for severe imbalance rather than as part of the main

Table 6: Average empirical size in the unbalanced-ratio diagnostic with $\rho_0 = 0.5$, $n_2 = 60$, $R = 1000$, and $B = 500$. Each row averages over the two covariance families and the three innovation distributions.

p	n_1	n_2	n_1/n_2	$G(0.3)$	$G(0.5)$	$G(0.7)$	I_p
100	300	60	5.0	0.117	0.111	0.098	0.057
100	180	60	3.0	0.095	0.097	0.090	0.050
100	120	60	2.0	0.065	0.065	0.061	0.035
100	90	60	1.5	0.057	0.060	0.059	0.030
100	60	60	1.0	0.050	0.057	0.057	0.027
200	300	60	5.0	0.105	0.101	0.092	0.054
200	180	60	3.0	0.089	0.089	0.083	0.048
200	120	60	2.0	0.063	0.064	0.061	0.030
200	90	60	1.5	0.057	0.055	0.055	0.029
200	60	60	1.0	0.054	0.055	0.054	0.028

calibration evidence.

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